Risk Management in Green Supply Chain Management (GSCM): A Case Study on Supplier and Commodity Sustainability Risks

By

ANNE MARIE TAN YUN XIN

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY
PENANG

ACADEMIC YEAR
2025/26

Risk Management in Green Supply Chain Management (GSCM): A Case Study on Supplier and Commodity Sustainability Risks

By

ANNE MARIE TAN YUN XIN

Supervisor: DR. LIM KHAI YIN

A project report submitted to the
Faculty of Computing and Information Technology
in partial fulfillment of the requirement for the
Bachelor of Computer Science (Honours)

Department of Computing and Information Technology
Faculty of Computing and Information Technology
Tunku Abdul Rahman University of Management and Technology
Penang

Declaration

The project submitted herewith is a result of my own efforts in totality and in every aspect of the project works. All information that has been obtained from other sources had been fully acknowledged. I understand that any plagiarism, cheating or collusion or any sorts constitutes a breach of TAR University rules and regulations and would be subjected to disciplinary actions.

Anne Marie Tan Yun Xin

Bachelor of Computer Science (Honours) in Data Science

ID: 24PMR10314

Abstract

This study introduces a data-driven framework for risk management in Green Supply Chain Management (GSCM), integrating predictive machine learning (ML), multi-criteria decision-making (MCDM), and generative modelling techniques. The framework is designed to address critical challenges, including the limited adoption of artificial intelligence (AI)-driven predictive analytics, restricted visibility into deeper-tier suppliers, and the insufficient integration of sustainability metrics in conventional supply chain risk management (SCRM) systems. Predictive models are developed to identify and mitigate supplier and commodity risks, while statistical methods quantify the influence of environmental factors, such as carbon and greenhouse gas (GHG) emissions on key GSCM decision variables. The fuzzy AHP-TOPSIS approach is optimised to ensure sustainability criteria remain central to decision-making, and generative AI (GenAI) models are employed to create realistic synthetic supplier and commodity data, enhancing visibility across the supply network. The findings indicate that the proposed framework offers actionable insights, enabling organisations to strengthen supply chain resilience while aligning with sustainability objectives. This research makes a significant contribution by bridging the gap between traditional risk management approaches and the growing need for environmentally responsible, AI-driven supply chain solutions.

Acknowledgement

I would like to express my heartfelt gratitude to my supervisor, Dr. Lim Khai Yin, whose guidance, encouragement, and expertise have been pivotal throughout the entire course of this project. Her insightful feedback and constructive criticism have challenged me to think critically, refine my ideas, and improve the overall quality of my research. I am especially grateful for her patience and unwavering support in guiding me through both the theoretical and practical aspects of this work, ensuring that I remained focused and motivated despite the challenges encountered.

My sincere thanks also go to my peers, who have been an invaluable source of collaboration and inspiration. The many discussions we had, whether related to research design, coding implementation, or report writing, provided fresh perspectives and practical solutions that enhanced the progress of this project. Their willingness to share resources, troubleshoot problems together, and offer encouragement created a positive and productive environment that made the journey more rewarding.

Finally, I would like to extend my deepest appreciation to my family for their constant support, understanding, and encouragement throughout this endeavour. Their patience during long hours of research and writing, their reassurance during moments of doubt, and their unwavering belief in my abilities have been my greatest source of motivation. Without their moral and emotional support, this project would not have been possible.

Table of Contents

Chapter 1

Introduction

1 Introduction

In Chapter 1, the foundation of this study is established by first presenting the background and context of Green Supply Chain Management (GSCM) and its relevance to sustainability. This is followed by problem statements that outlines the key challenges in GSCM risk management. This chapter then specifies the research objectives and defines the scope of the study to set precise boundaries. Finally, it highlights existing competitive solutions and explains the unique contributions of the proposed framework.

1.1 Background

In recent years, the global supply chain landscape has significantly shifted towards adopting greener efforts, with industries facing mounting pressure to operate sustainably while maintaining efficiency and resilience, as well as adhering to their company's values and protocols. Environmental concerns, climate regulations, and growing consumer awareness have pushed companies to restructure their operational strategies, particularly in how they source, manufacture, and distribute products. This sudden surge in green and sustainable practices has resulted in the emergence of Green Supply Chain Management (GSCM), which aims to minimise environmental impacts and respective carbon footprint, whilst ensuring long-term competitiveness between industries. This can be seen in Figure 1.1(a) — 2024 Statista report showing over 92% of companies in the Asia-Pacific region have partaken sustainability reporting, indicating the rising demand for transparency and environmentally responsible supply chain practices (Statista Research Department, 2024). Moreover, in a 2023 Statista report, it was found that supply chain disruptions (59%) were one of the main drivers in Information Technology (IT) for the emphasis on improving sustainability worldwide, as observed in Figure 1.1(b) (Sujay Vailshery, 2024).

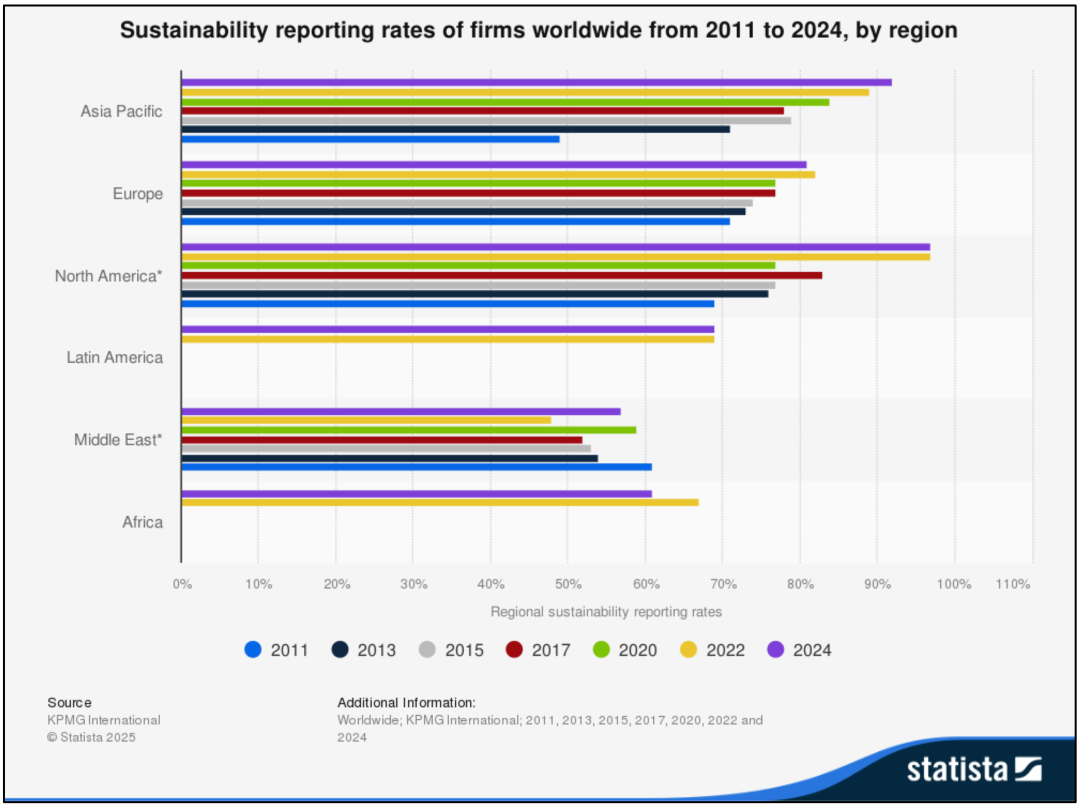

Figure 1.1(a): Sustainability reporting rates of firms worldwide from 2011 to 2024, by region (Statista Research Department, 2024).

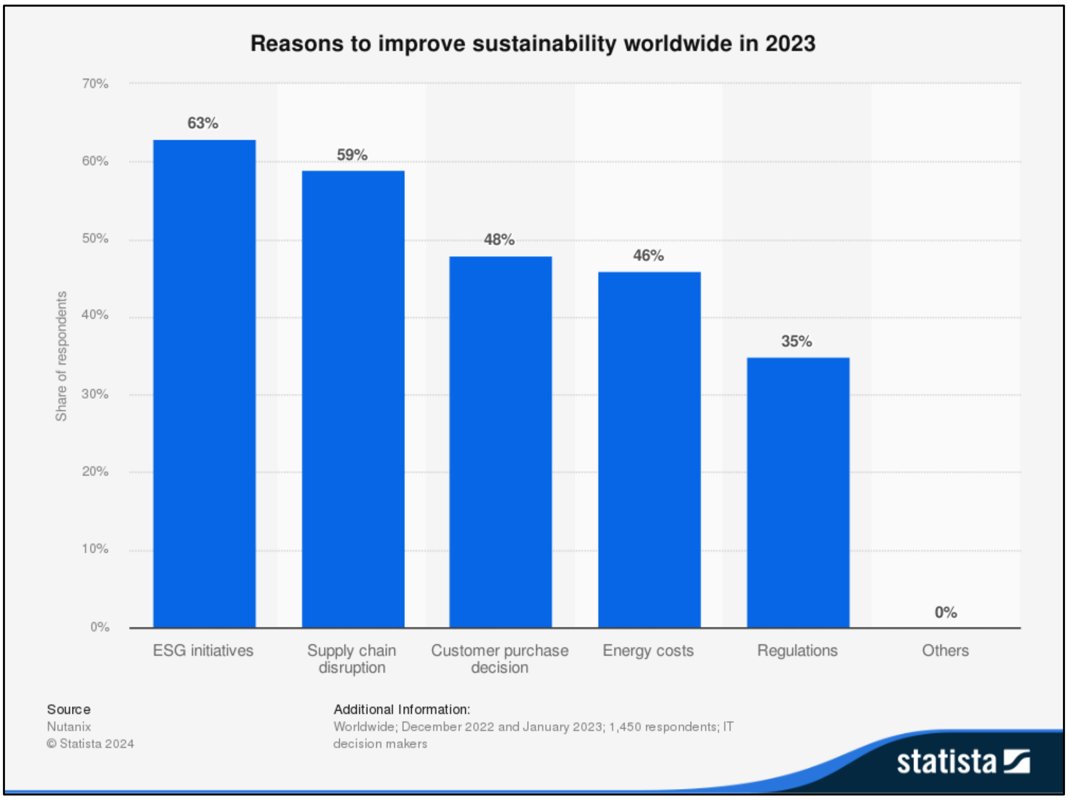

Figure 1.1(b): Reasons to improve sustainability worldwide in 2023 (Sujay Vailshery, 2024).

Notably, many companies continue to struggle with identifying and mitigating sustainability-related risks, such as carbon emissions, resource inefficiency, and supplier intransparency and non-compliance. This is largely due to the limited integration of AI-driven predictive risk management tools that consider environmental sustainability metrics. Authors Zejjari and Benhayoun (2024) have recognised the missed opportunities of the implementation of AI for sustainable supply chain management, where they emphasised the limited attention to and effort invested into Sustainable Development Goals (SDGs) in supply chain operations, despite the rise in AI-driven risk management solutions catered specifically for GSCM. Traditional SCRM frameworks primarily focus on financial and operational risks, often overlooking critical green factors. This arrangement has made it difficult for business to align with SDGs and achieve carbon neutrality. Even where companies have implemented SCRM tools, many remain uncertain whether these systems are optimised to meet GSCM standards. Dwidienawati et al. (2025) echoed concerns about the lack of employee awareness around core sustainability principles. The authors also found out that managers often blindly implement eco-friendly practices without fully recognising their GSCM significance. Additionally, the lack of knowledge on how carbon and GHG emissions influence supply chain decisions are persistent in many companies. Finally, the scarcity of publicly available supplier data exacerbates transparency challenges and hinders trustworthy partnership formation, as documented in studies of Jia et al. (2024).

Given the global shift toward sustainability and the increasing complexity of modern supply chains, the proposed research is highly relevant and timely. As consumers and stakeholders shed light on sustainability standards and green supply chain practices, businesses must not only comply with sustainability standards, but also proactively manage risks associated with GSCM. Even now, current risk management system often fails to account for sustainability metrics, like carbon and GHG emissions, supplier environmental compliance, and resource efficiency. This scenario leaves organisations vulnerable to reputational, regulatory, and operational risks. Thus, by introducing an AI-driven risk management framework specifically designed for GSCM, this research addresses a critical gap in existing systems. It integrates ML, optimisation algorithms, and GenAI to improve the accuracy of sustainability risk assessment, even in the face of limited supplier data. The outcome has the potential to empower businesses with better decision-making tools, contribute to SDGs, and foster more resilient, transparent, and environmentally responsible supply chains, making this study not only valuable but necessary.

1.2 Problem Statement

As the global market progressively shifts towards sustainability, companies are actively seeking optimal solutions to integrate environmentally responsible practices into their operations. GSCM has become a key focus across industries and has driven the need to consider and incorporate 'green' factors, such as carbon footprint reduction, resource efficiency, and supplier sustainability compliance. Nonetheless, the transition to greener supply chains comes with significant risks, including supplier intransparency and operational disruptions. Therefore, these challenges highlight the urgent call for robust risk management strategies to ensure sustainability efforts will be able to optimise supply chain resilience and efficiency.

In 2021, Statista reported that while 22% of supply chain companies had implemented predictive and prescriptive analytics into their company operations, a significant 82% only planned to do so within the next 5 years. Similarly, 14% had adopted AI technologies, with 73% intending to integrate them within the same timeframe (Placek, 2024). Despite the advancements in technology, these statistics have shown that there is limited adoption of predictive modelling with AI and ML integration in GSCM. The lack of predictive modelling tools for sustainability risks in supply chains impede efficient decision-making, as well as hinders companies from aligning their operations with SDGs.

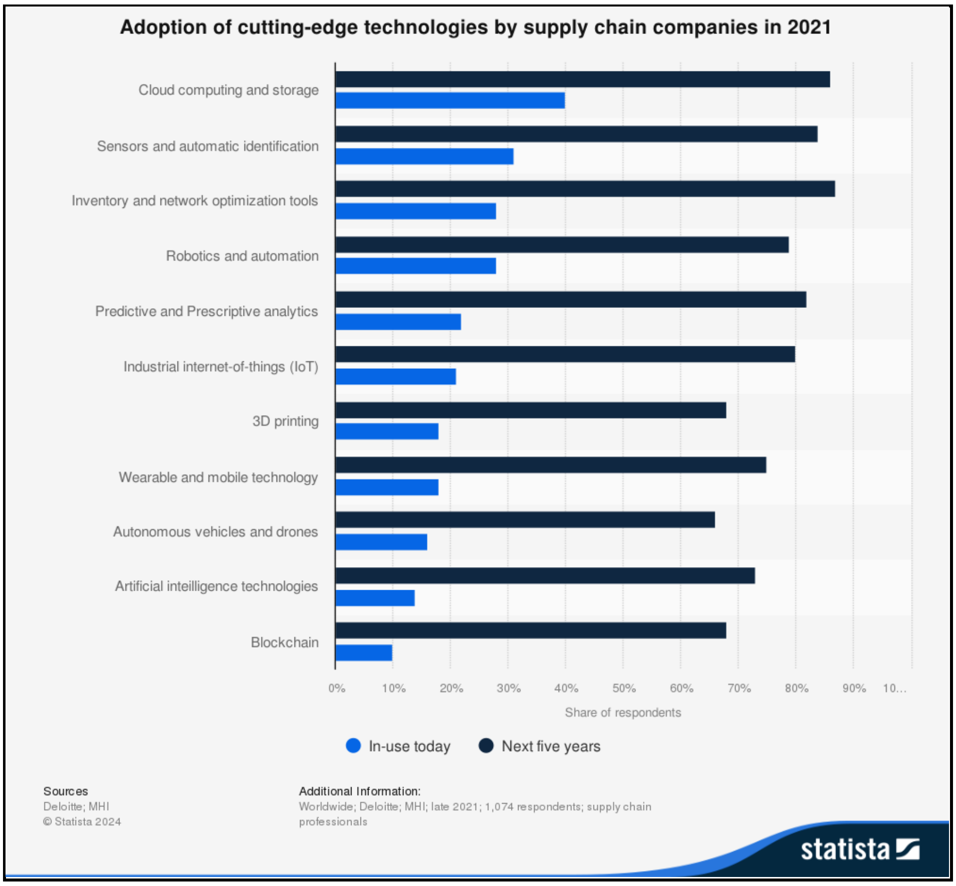

Figure 1.2: Adoption of cutting-edge technologies by supply chain companies in 2021 (Placek, 2024).

Notably, many companies do implement SCRM, albeit unaware that they are not fully optimised to meet GSCM standards. Two commonly used algorithms in SCRM were fuzzy AHP and fuzzy TOPSIS, with some studies proposing a hybrid version of the two, or fuzzy AHP-TOPSIS (Asih et al., 2025;Gupta et al., 2025;Varchandi et al., 2024;Zulqarnain et al., 2024;Majumdar et al., 2023).This algorithm can perform well under normal SCRM circumstances but often fails to account the "green" factor into their frameworks (Dhumras & Bajaj, 2024). These findings suggest that companies are aware of GSCM but have yet to actively implement them properly with its causality being perceived cost, perceived relative advantage, perceived complexity, compatibility, quality of human resources, top management support, company size, environmental uncertainty, regulatory pressure, and customer pressure (Lin et al., 2020).

Furthermore, some companies unfairly balance economic performance and environmental responsibility and prioritise cost efficiency over sustainability. This roots from the uncertainty in decision-making, where environmental factors like carbon and GHG emissions are unclear as to how they directly or indirectly influence the operations in GSCM. A study examined the relationship between GSCM, digital transformation, and corporate performance, and strongly emphasized that companies pursuing carbon neutrality must consider carbon and GHG emissions into strategic decision-making technology adoption and operations (Y. Chen & Guo, 2024). Therefore, the importance of transparency and awareness in sustainability efforts cannot be exaggerated, as the lack of crucial sustainability insights may hinder companies from making informed decisions as they strive towards implementing GSCM. Additionally, companies must evaluate the influence of carbon and GHG emissions to enable long-term environmental and economic sustainability.

Another reason why it is challenging for SCRM models to accurately identify sustainable practices in suppliers is the lack of publicly available supplier data. Understandably, many suppliers choose not to disclose key information related to their sustainability efforts, compliance records, and operational risks due to concerns over confidentiality. However, this lack of transparency has impeded companies to identify potential risks while ensuring compliance with sustainability standards in supply chain management (Whieldon et al., 2024). A study found that most companies only track their tier-1 suppliers, while data on deeper-tier suppliers, mainly tier-2 and tier-3 suppliers, remained intransparent, incomplete, or entirely unavailable. In fact, the authors found that 80% of companies could not name their tier-2 and tier-3 suppliers (Liu et al., 2023),thus demonstrating the importance of supply chain visibility beyond direct suppliers in assessing risks and implementing green practices effectively.

1.3 Objectives

This study sets out four principal objectives, each aimed at addressing the problem statements outlined in Section 1.2.

1. **To develop predictive ML models for risk identification and mitigation in GSCM by using Random Forest (RF), XGBoost, and Artificial Neural Network (ANN).**

   This objective directly tackles the limited adoption of predictive modelling and AI integration in GSCM by introducing a data-driven framework that leverages state-of-the-art machine ML algorithms. By developing predictive models using RF, XGBoost, and ANN, this study aims to identify and quantify supplier and commodity sustainability risks in a systematic and reproducible manner. These algorithms will be used for handling high-dimensional, heterogeneous supply chain data, capturing complex, non-linear relationships between risk factors and outcomes.

2. **To quantify the relationship between carbon and GHG emissions, alongside other risk factors with specific GSCM decision variables using Multiple Linear Regression (MLR).**

   This objective addresses the lack of transparency and understanding of how carbon and GHG emissions influence decision-making in GSCM by quantifying their relationship with key operational variables using MLR. By modelling dependent decision variables, like supplier selection and sustainability scores against independent risk factors such as carbon intensity, GHG emissions, and regulatory pressures, MLR can estimate the magnitude and direction of each factor's impact while controlling other variables. This provides a clear understanding of how environmental metrics interact with economic and operational outcomes.

3. **To generate supplier information, as well as additional necessary information for commodities with generative AI models, such as Conditional Tabular Generative Adversarial Network (CTGAN) and Tabular Variational Autoencoder (TVAE).**

   This objective addresses the critical challenge of limited supplier transparency by leveraging GenAI models such as CTGAN and TVAE to create realistic, statistically consistent synthetic data that represent supplier characteristics, sustainability indicators, and commodity-related information. By training these models on the available tier-1 supplier and commodity data, it becomes possible to infer and generate plausible data for deeper-tier suppliers like tier-2 and tier-3, filling the gaps left by incomplete or undisclosed records. This approach aims simulate supply chain scenarios, identify potential risk points, and evaluate supplier compliance even when real-world data is scarce or inaccessible due to confidentiality concerns. Moreover, synthetic data generated in this way preserves the underlying statistical distribution of the original data while mitigating privacy risks, enabling robust analysis without compromising sensitive information.

1.4 Research Scope

This study will use two primary datasets and one synthetically generated dataset to support the development of the proposed GSCM risk management framework. The first dataset, *ImportExport Trade data Asia.csv*, captures international trade activities across Asia and will be used to analyse commodity flow, trading patterns, and potential sustainability risks associated with specific products or regions. The second dataset, *SupplyChainGHGEmissionFactors_v1.2_NAICS_byGHG_USD2021.csv*, provides detailed emission factors by industry classification (NAICS), enabling the quantification of carbon and GHG emissions linked to traded commodities. These emissions data will be essential in assessing environmental impact and guiding decision-making in sustainable supply chain management. Due to the limited availability of real-world supplier sustainability data, this study will also generate synthetic supplier data using Conditional Tabular GAN (CTGAN) and Tabular Variational Autoencoder (TVAE). This artificial dataset will include key variables such as ESG scores, compliance indicators, lead time, and production capacity, thereby enabling comprehensive supplier risk analysis despite real-world data limitations. These datasets will be used together provide a robust foundation for predictive modelling, clustering, and optimisation within the proposed AI-driven risk management framework.

To address the problem statements, a predictive ML model for risk management in GSCM will be developed with RF, XGBoost, and ANN. These models will be trained to identify potential risks by analysing trade and emissions data, along with supplier-related information. Prior to model development, feature selection will be conducted to determine the most influential risk factors in commodity and supplier sustainability. Additionally, suppliers and commodities will be grouped based on risk levels with clustering techniques like DBSCAN and K-Means. Within each cluster, the risk threshold for the three risk groups (low risk, moderate risk, and high risk) will be determined using interquartile range (IQR). For instance, the following outline shows how risk levels would be assigned based on their percentiles:

- Low Risk : Below Q1 (25th percentile).
- Moderate Risk : Between Q1 and Q3 (25th - 75th percentile).

- High Risk : Above Q3 (75th percentile).

Next, Multiple Linear Regression (MLR) will be used to quantify the relationship between carbon and GHG emissions, alongside other risk factors with specific GSCM decision variables. As mentioned previously, carbon and GHG emissions need to be embodied into decision-making technology adoption and operations of companies pursuing carbon neutrality. Therefore, this approach will help identify the extent to which carbon and GHG emissions influence key decision-making processes in GSCM.

Additionally, the existing risk management framework — fuzzy AHP-TOPSIS will be optimised with GA. Specifically, fuzzy AHP will be optimised for weightage assignment, whereas fuzzy TOPSIS will be optimised for ranking suppliers and commodities. The ability of GA to efficiently explore a vast search space and identify optimal weight distributions makes it ideal for handling complexity and uncertainty of sustainability risk factors in fuzzy AHP. On the other hand, it will also be used in fuzzy TOPSIS to fine-tune ranking solutions by iteratively enhancing risk prioritisation based on predefined criteria. These optimisation techniques serve to optimise the overall effectiveness of current GSCM risk management frameworks in terms of accuracy and reliability by providing a more precise evaluation of sustainability factors, reducing subjectivity and inconsistencies in decision-making.

Lastly, to address insufficient and incomplete supplier data, GenAI models such as CTGAN and TVAE will be used to generate synthetic supplier information. Supplier information like Environmental, Social, and Governance (ESG) scores, supplier financial stability, supply chain compliance and certifications, and sustainability reports on emission levels and waste management need to be included, alongside other operational factors like lead time, production capacity, and historical performance to maximise the accuracy and reliability of GSCM risk management models. Moreover, these models will also be leveraged to generate any necessary missing information for commodities. This synthetic data serves as a representative benchmark to enable a more informed and data-driven approach to sustainable supply chain decision-making.

1.5 Competitions and Contributions

The primary competition to the proposed GSCM risk management framework consists of existing AI-powered SCRM platforms designed to enhance operational resilience and mitigate disruptions. Notable competitors include QIMA (2025), which focuses on ESG compliance monitoring and risk mitigation, and Pecan AI (2025) and interos.ai (2025), which deliver ML-powered insights into supplier performance, operational risks, and global supply chain dependencies. Additionally, SAP (2025) is a global leader in enterprise software, provides sophisticated supply chain solutions such as SAP Integrated Business Planning (IBP) and SAP Business AI, which incorporate predictive analytics, anomaly detection, and optimisation models to improve demand planning, inventory management, and supplier risk assessment.

Despite their capabilities, these solutions exhibit a significant gap in addressing decision-making in terms of sustainability. While they successfully enhanced supply chain efficiency and resilience, they rarely prioritise GSCM metrics. This is critical, as Statista reported that 34% of supply chain and manufacturing businesses had adopted AI widely by 2022, with 11% reporting it as critical for growth, and 38% predicting it will be essential by 2025 (Thormundsson, 2023). The figures highlight an increasing recognition of AI's importance but also underscore the untapped potential of AI-driven sustainability modelling.

Unlike traditional SCRM platforms, the proposed GSCM risk management framework will leverage ML-driven predictive modelling with an optimised fuzzy AHP-TOPSIS for sustainability-centered decision-making. Moreover, the role of GenAI in generating synthetic supplier data will facilitate risk assessment for environmentally responsible supply chains. This innovation will cater specifically to industries facing regulatory pressures for carbon neutrality, such as automotive manufacturing, consumer electronics, and fast-moving consumer goods (FMCS), where sustainable sourcing is crucial.

In short, while current competitive solutions like QIMA, Pecan AI, interos.ai, and SAP SE focus mainly on SCRM, this study targets a sustainability-focused optimization on SCRM that contributes to bridging the gap between conventional risk management tools and the increasing demand for sustainable supply chains. This model will provide businesses with an overview and the possibility of enhancing the resilience of their supply chains while meeting "green" objectives, making it a unique contribution to the field of AI-driven GSCM risk management.

1.6 Chapter Summary and Evaluation

Chapter 1 establishes the foundation of the study by outlining the background of GSCM, defining the problem statement, and identifying the research objectives. It specifies the scope of the study and positions the proposed framework within the context of existing solutions by comparing it with other AI-powered SCRM platforms. This chapter justifies the need for the research and clearly articulates this study's contributions, ensuring alignment between the identified problems and the proposed solution.

Chapter 2

Literature Review

2 Literature Review

In Chapter 2, existing studies relevant to GSCM risk assessment were reviewed. This chapter covers sustainability integration in supply chains, AI and ML applications for predictive risk modelling, and the use of MCDM methods for supplier and commodity evaluation. It also explores synthetic data generation to address data scarcity and examines clustering and feature selection techniques for risk profiling. The chapter concludes with a discussion that identifies research gaps and positions the proposed framework within the current body of knowledge.

2.1 Existing Studies

2.1.1 Studies on Sustainability Integration in Supply Chain Management

A study by Singh et al. (2025) investigated how digital transformation (DT) initiatives influence green supply chain performance (GSCP), particularly in achieving carbon neutrality and supporting circular economy (CE) goals. Here, the investigation adopts a theoretical model established on a resource-based, data empowerments, and contingency theory. The authors have conducted a qualitative research design, collecting data via surveys from 350 IT professionals in India, where 310 valid questionnaires were analysed using Partial Least Squares Structural Equation Modelling (PLS-SEM). This research aimed to examine how green supply chain integration (GSCI) mediates the relationship between DT and GSCP, along with the absorptive capacity and economic policy uncertainty (EPU) as moderators. Key findings from the study indicated that DT significantly enhances GSCP when GSCI is effectively implemented, and absorptive capacity positively moderates this bond. Conversely, EPU did not exhibit a significant moderating effect between GSCP and GSCI. Overall, this study emphasises the importance of digital innovation, organisational learning capacity, and environmental adaptability in enhancing GSCM. These insights reinforce the value of integrating DT and AI-driven strategies into sustainability frameworks, especially for organisations striving to improve their environmental performance through GSCM.

In another study by Alabdali et al. (2025) the authors explored how green digital transformational leadership (G-DTL) affects green digital supply chain transformation (G-DSCT) and resilience (G-DSCR) through two serial mediating variables, namely algorithmic management (ALGM) and digital absorptive capacity (G-DAC). The proposed serial mediation model was drawn on top of the transformational leadership theory, absorptive capacity theory, and the stimulus-organism-response framework. In this study, survey data from 324 supply chain professionals in Saudi Arabia were collected and analysed using the PLS-SEM method. As a result, it was concluded that G-DTL had a strong positive impact on both G-DSCT and G-DSCR, and that this relationship was significantly mediated in sequence by ALGM and G-DAC. These results underscore the importance of digital leadership competencies, AI-based management tools, and organisational learning in fostering sustainable and resilient supply chains. Similar to the aforementioned study by Singh et al. (2025), this study reinforces the idea that DT alone is insufficient in achieving effective green supply chain outcomes, rather it must be strategically led and supported by technological and absorptive capabilities. Both studies have provided strong empirical evidence that digital integration and organisational adaptability are essential drivers of successful GSCM.

Furthermore, a study titled *"A literature review on green supply chain management for sustainable sourcing and distribution"* by Hariyani et al. (2024) presents a comprehensive review of GSCM practices, focusing on the domains of sustainable sourcing and distribution. With the growing environmental awareness in various industries, this study was dedicated to shed light on core GSCM principles, distinguish between sourcing and distribution practices, examine theoretical frameworks that control and influence GSCM practices, assess recent innovations, and identify barriers such as financial, technological, and regulatory challenges. The authors have extracted 266 peer-reviewed papers published from 1997 to 2024 by utilising the Scopus database and the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) methodology. From the study, the authors found that sustainable sourcing was commonly enabled by technologies like AI and blockchain to ensure traceability, material innovation, such as bioplastics made from renewable sources, and supplier collaboration through training and incentives. Meanwhile in distribution, green logistics, like electric vehicles (EVs) and route optimisation, and energy-efficient warehousing play critical roles. In summary, this study provides broader insights into the conceptual and technological landscape of GSCM while reinforcing the importance of innovation, DT, and stakeholder engagement for adopting sustainability in operations.

Moreover, Junejo et al. (2025) has conducted a study to investigate the influence of GSCM on the sustainable performance of small and medium enterprises (SMEs) in Pakistan. The authors specifically examined the mediating roles of green knowledge sharing (GKS), green innovation (GI), and big data-driven supply chains (BDDSC). In this study, data was collected through questionnaires from 469 SME employees and analysed using SEM via SmartPLS 3. The findings revealed that through direct mediators GKS, CI, and BDDSC, a significant positive relationship between GSCM and sustainable performance (SP) was observed. Overall, this study emphasises that in developing countries, support for innovation, knowledge dissemination, and digital technologies like big data are crucial for implementing sustainable practices. This study underscores the relevance of contextual factors in emerging economies and provides actionable insights for policymakers and SME managers to adopt and strengthen sustainability in supply chains.

In another research titled *"Flexible Green Supply Chain Management in Emerging Economies: A Systematic Literature Review"*, the authors have leveraged a mixed methodology of systematic literature review, text mining, and network analysis to evaluate 108 peer-reviewed articles published between 2000 and 2021. The study applied the contingency theory and institutional theory to explore flexible green supply chain management (FGSCM) strategies, practices, drivers, and challenges. After the investigation, it was revealed that FGSCM research in emerging economies was still in its infancy compared to developed countries. Additionally, the review highlighted significant research gaps, including limited contextualised frameworks and the separation of flexibility and environmental sustainability in supply chain research. Notably, the authors had proposed a research framework for FGSCM, which has been tailored to the prominent pressures of emerging economies. Those said pressures include coercive pressure, which concerns whether regulatory rules have been complied with or not, mimetic pressure, which concerns cultural cognitive pressures, and normative pressure, which is whether there were in response to customer pressure. In summary, this study supports the need for adaptability in localised GSCM models under environmental and operational uncertainties, especially in regions where industrialisation rapidly grows (Dhillon et al., 2022).

Balkumar et al. (2024) also contributed to this area by providing a thorough analysis of global research trends in GSCM. Through this systematic literature review, the authors have categorised and assessed the evolution of GSCM topics, highlighting growing academic and industrial interest in green purchasing, reverse logistics, eco-design, and life-cycle assessment. Though not limited to a specific region, the paper mapped out thematic clusters and identified an incline towards integrating digital technologies, stakeholder collaboration, and sustainability performance metrics in supply chain operations. One notable finding was the growing relevance of GSCM in corporate strategy, moving beyond regulatory compliance to competitive differentiation. Similar to studies by Junejo et al. (2025) and Dhillon et al. (2022) these authors have voiced an urgency for industries to develop data-driven and adaptive green supply chains while aligning both environmental goals and business outcomes.

In a related study titled *"Green Supply Chain Management in Manufacturing Firms: A Resource-Based Viewpoint"*, the resource-based view (RBV) theory has been applied to examine how internal organisational resources contribute to the successful implementation of GSCM in manufacturing firms. The study surveyed 381 employees from UK-based manufacturing companies and used SEM to analyse the relationships among key factors, including supply chain connectivity (SCC), supply chain information sharing (SCIS), top management commitment (TMC), and green procurement and logistics acceptance (GPLA). The authors revealed that SCC and SCIS positively influence TMC and GPLA, which in turn directly and strongly affected the effectiveness of GSCM implementation. Moreover, SCC and SCIS were found to have indirect effects on GSCM through the serial mediation of TMC and GPLA. A notable contribution of the study was the development and validation of multidimensional GSCM measurement scale rooted in the RBV theory. The scale offered a theoretically robust and empirically tested framework for assessing GSCM practices within organisations, especially in the context of manufacturing firms. Unlike many prior studies that focused on fragmented or single-dimensional assessments of green practices, this scale integrated the aforementioned critical components into a cohesive model. Additionally, it has the ability to capture tangible and intangible organisational resources that influence GSCM. Overall, this study provides a broader discussion, emphasising that external technological enablers alone are insufficient without the supporting organisational infrastructure and leadership to integrate and sustain innovations within green supply chain initiatives (Khan et al., 2023).

In a complementary research by Yang et al. (2024), the authors focused on understanding how sustainability is embedded within service supply chains (SSCs), which was a relatively underexplored area compared to product-oriented supply chains. Using the PRISMA framework, the authors systematically reviewed 87 peer-reviewed journal articles published across top logistics and supply chain journals, such as the International Journal of Physical Distribution & Logistics Management, the International Journal of Production Economies, and so on. The objective of this study was to identify the primary drivers, enablers, and barriers that influence sustainable practices in SSCCs, and to propose a conceptual framework that facilitates future research in this domain. At the end of the study, the authors found that technological enablers, such as AI, IoT, and digital platforms, as well as stakeholder pressure, institutional influence, and organisational culture were essential in facilitating the integration of environmental and social considerations into the service-based supply chains. In addition to that, this paper expanded the sustainability discourse by emphasising that although services may be intangible, they play a critical role in achieving broader environmental goals, particularly via green service design, responsible outsourcing, and eco-efficient logistics. In conclusion, this paper has voiced the significance of DT, leadership, and organisational adaptability as universal enablers of sustainability, which are not limited to the manufacturing and product-based supply chains, but also in the increasingly important service sector. Yang et al. (2024) and Khan et al. (2023) alike have echoed the crucial role of organisational enablers in successfully integrating sustainability into supply chain practices. Both studies have pointed out that digital or technological tools must be supported by strategic internal capabilities to generate meaningful and lasting sustainability outcomes in supply chains.

An additional investigation by Atieh & Abushaega (2025) was conducted to examine how internal dynamic capabilities, namely digital leadership, environmental awareness, and organisational learning can affect green supply chain innovation (GSCI), and sustainable supply chain performance (SSCP). Grounded in the Dynamics Capabilities Theory (DCT), the authors developed a conceptual model and tested it using data from 312 logistics and supply chain professionals in a Middle East country called Jordan. The model integrated all three capabilities within the logistics sector of an emerging market in Jordanian markets, offering valuable insights for firms operating under resource constraints. By utilising PLS-SEM, the study confirmed that all three capabilities influence GSCI, which in turn strongly drives SSCP. Notably, digital leadership emerged as the strongest predictor of GSCI with $\beta = 0.481$, followed by organisational learning with $\beta = 0.313$ and environmental awareness with $\beta = 0.151$. The mediating role of GSCI was also confirmed, demonstrating that innovation is a critical mechanism through which internal capabilities translate into sustainable outcomes. These findings reinforce the growing relevance of technology-driven, learning-oriented, and environmentally aware strategies in driving green innovation and achieving sustainability in supply chains.

Building on this, Kholaif et al. (2024) published *"Post-emergencies opportunities for green supply chain management and urban supply chains' robustness; the moderating influence of blockchains as a sustainable urban resources management technique"* to explore how green supply chain practices and blockchain technologies (BCT) can strengthen the robustness of urban supply chains, especially for SMEs in Egypt during post-pandemic recovery. The study addresses the impact of uncertainty and anxiety stemming from crises like COVID-19 on supply chain structures, emphasising the shift in SMEs toward sustainable practices in response. The authors used data collected from 417 SME managers in Egypt, and analysed them via the PLS-SEM method while examining several hypotheses. They found that the pandemic had positively induced uncertainty into the adoption of green supply chain practices and that BCT moderated this relationship by enhancing supply chain transparency, trust, and resilience. Notably, green supply chain practices were also shown to mediate the relationship between pandemic uncertainty and urban supply chain robustness, suggesting that green initiatives can serve as a buffer in times of disruption. Another unique element of this study is that it is an integration of several theories, like the Social Cognitive Theory, DCT, Crisis Management Theory, and Supply Chain Resilience Theory. Overall, this study stresses and supports the argument for AI and technology-driven sustainable supply chain frameworks in developing economies.

2.1.2 AI and ML Applications in GSCM Risk Management

A research paper titled *"Risk Analysis on Green Supply Chain Management: A Systematic Literature Review"* by Jayasinghe et al. (2023) has offered a detailed synthesis of research conducted between 2007 and 2022 on the risks associated with implementing GSCM practices. The authors had performed a systematic literature review of 30 selected studies filtered from Google Scholar, Science Direct, and ResearchGate, aiming to categorise the types of risks encountered, highlight the industries studies, and evaluate the analytical methods used. In this study, critical risk categories, such as manufacturing, supply, information technology (IT), financial, environmental, and social risks are identified, as well as emphasising that risk priorities vary significantly across industries and countries. Notably, the authors used quantitative methods for risk analysis, particularly MCDM models like fuzzy AHP, fuzzy TOPSIS, fuzzy failure mode and effect analysis (FMEA), fuzzy decision making trial and evaluation laboratory (DEMATEL), and hybrid approaches integrating these with simulation or interpretive ranking processes. These methods were commonly used to develop vulnerability matrices, benchmark models, and risk prioritisation frameworks. Additionally, the authors mapped risk types industry-by-industry along with their severity. This has illustrated how digital and AI-driven tools must be context-sensitive when deployed in GSCM. All in all, this study reinforces the importance of advanced decision-making and AI-enhanced risk analytics for a sustainable supply chain management.

A related study by Nezianya et al. (2024) contributed a comprehensive and structured evaluation of how ML is reshaping traditional SCRM. The objective of this study was to highlight the potential of ML in enhancing supply chain resilience through risk prediction, real-time visibility, supplier evaluation, and fraud detection. Unlike traditional risk mitigation strategies, which rely heavily on historical data and manual decision-making, the research underscores how ML models process large, diverse datasets to identify patterns, forecast disruptions, and optimise inventory and supply performance. The methodology employed is a systematic literature review of recent academic and industry sources, which were selected based on inclusion criteria. For example, studies that discussed and focused on ML applications in SCRM were selected for this research. At the end of the research, the authors had identified severely kep ML applications, including demand forecasting using neural networks (NN), inventory optimisation via predictive algorithm, and supplier risk assessment through continuous data monitoring. They also discussed the integration of ML with real-time visibility tools and simulation-based scenario analysis. Despite its benefits, the paper acknowledges substantial challenges, like data quality issues, integration of ML approaches with legacy systems, high implementation costs, scalability issues, and organisational resistance. In conclusion, the article argues that while ML has strong potential to transform SCRM from reactive to proactive, future research should focus on improving data governance, ethical AI practices, and technology integration to fully realise the benefits of ML in SCRM.

Furthermore, a comprehensive literature review provided by Gidiagba et al. (2025) examines how ML techniques are being applied in sustainable supplier selection (SSS) from 2010 to 2024. The primary objective of the study is to evaluate the evolution, methods, challenges, and future directions of ML in supplier selection, while focusing on sustainability aspects like environmental impact, social responsibility, and economic viability. Using a systematic literature review methodology alongside the PRISMA framework, the authors filtered and sessed a total of 50 peer-reviewed papers from databases like Scopus, Web of Science, and Google Scholar. Results show that the ML models under discussion — Support Vector Machine (SVM), Artificial Neural Network (ANN), Random Forest (RF), and hybrid models — combined with MCDM were able to enhance precision, efficiency, and reliability of supplier evaluation compared to traditional methods. These models are especially effective in processing large and complex datasets, and adapting to dynamic supply chain environments. Additionally, this study has identified gaps in SSS, such as limited use of ML in certain sectors, challenges with data quality and model interpretability, and a need for integration with other emerging technologies like IoT and blockchain. In this study, the authors have performed cross-sectoral comparisons, policy implications for ML adoption, and ethical considerations in the use of AI for supplier selection. The authors have also concluded that ML holds transformative potential for SSS and advocated for more integrated, context-sensitive, and ethically aligned approaches in the future.

In a separate study titled *"Green supply chain transformation and emission reduction based on ML"*, authors Wu & Zuo (2023) presented a theoretical investigation into how ML-driven carbon emission reduction technologies influence strategic decisions of competing supply chinese. Using a Cournot duopoly model, the authors developed two frameworks, mainly one under symmetric information and another under asymmetric information to simulate how uncertainty about ML upgrade success affects production output, pricing strategies, and market equilibrium. Notably, the authors took an interesting approach by applying game theory and comparative statics, rather than empirical datasets to evaluate the impact of ML investment risks. The findings revealed that under symmetric information, the risk associated with ML upgrades does not alter competitive equilibrium. Conversely, uncertainty significantly distorts firm behaviour, output levels, and pricing under asymmetric conditions. This demonstrates that ML technology adoption is not merely a technical upgrade but a strategic maneuver influenced by informational transparency. Thus, the authors advocated for subsidies and financial support backed by the government in stabilising investment incentives, as well as widespread adoption of ML in supply chains.

Moreover, M. Yang et al. (2023) a comprehensive review of the intersection between ML and SCRM. The primary objective of the study is to explore the current state of ML applications in SCRM, identify research gaps, and propose future research directions. The authors analysed 67 peer-reviewed articles sourced from nine major academic databases, using systematic selection criteria and bibliometric tools such as VOSviewer. The review categorises the ML applications into four core SCRM processes, that are, risk identification, risk assessment, risk mitigation, and risk monitoring. The key ML algorithms highlighted include deep learning, NN, SVMs, decision trees, and ensemble learning models. At the end of the review, findings revealed that while ML holds great potential in enhancing the efficiency and responsiveness of SCRM, especially under disruptions like the COVID-19 pandemic, adoption remains limited due to a significant interdisciplinary knowledge gap. Additionally, the study emphasises the need for more interpretable ML models, better integration of external data sources, and the incorporation of emerging technologies such as blockchain.

An additional study titled *"Designing and Deploying AI Models for Sustainable Logistics Optimization: A Case Study on Eco-Efficient Supply Chains in the USA"* by Shawon et al. (2025)

explored the transformative potential of AI and ML in optimising logistics operations with a focus on sustainability. In this study, the authors applied models, such as Linear Regression, XGBoost, SVMs, NN, K-Means, and DBSCAN on real-world logistics datasets collected from company databases and government transport records. This was to align with their goal to reduce carbon emissions, fuel consumption, and logistics costs. These models were used for multiple tasks, mainly demand forecasting, route optimisation, outlier detection, and clustering. Then, these models were evaluated with metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2. Notably, XGBoost outperformed other models in carbon emission prediction with an R^2 of 0.999. Meanwhile, the study also introduced hybrid deployment strategies which combined cloud and edge computing for real-time use. Moreover, fuel consumption, transport mode, especially air cargo, and elevation were found to be crucial predictors of carbon emissions. Overall, the study has underscored the effectiveness of AI in enabling eco-efficient logistics, while acknowledging similar barriers as mentioned by Nezianya et al. (2024) and Gidiagba et al. (2025), as well as lack of standardised sustainability metrics. It has been demonstrated that AI can significantly enhance logistics sustainability and operational efficiency in complex supply chain environments.

Further evidence is provided by Culot et al. (2024) who performed a systematic literature review of 123 empirical studies on the implementation of AI in supply chain management. The authors aimed to provide an evidence-based overview of how AI is shaping supply chain management, focusing particularly on the manufacturing sector. To avoid anecdotal evidence and theoretical bias, they exclusively analyse empirical studies sourced from peer-reviewed journals between 2010 and 2024, using Scopus and Web of Science databases. Their methodology involves rigorous screening and inductive content analysis using MAXQDATA software. In this study, four major thematic areas have been identified, namely, data and system requirements, such as the necessity for high-quality, large-scale data and the challenges of organisational data-sharing, technology deployment processes, including strategic alignment, resource allocation, and iterative AI model design, (inter)organisational integration, addressing workforce adaptation, trust issues, and structural changes, and finally, performance implications, particularly AI's benefits for operational efficiency, risk management, sustainability, and decision-making.The study also emphasises contextual factors like firm size, industry, and regulatory environments that influence AI adoption. Overall, the study concludes that while AI offers transformative potential in SCM, its successful implementation requires overcoming significant organisational, technical, and cultural hurdles, and calls for more targeted empirical research to refine integration strategies.

Elsewhere, Kassa et al. (2023) provided a thorough exploration of how AI technologies can be leveraged to build and support supply chain resilience (SCRes). The study focused on bridging fragmented literature by evaluating 106 peer-reviewed articles published between 2010 and 2022, focusing on the intersarion of AI and SCRes. The authors have adopted a systematic literature review methodology using a five-step PRISMA-guided process to synthesise and classify findings based on AI techniques, their capabilities, realised antecedents of SCRes, and the phases of resilience they support. In this study, Bayesian Networks, Multi-Agent Intelligent Systems (MAIS), ANNs, and genetic algorithms (GAs) were identified as the most utilised AI methods in resilience-building tasks, such as supplier selection, disruption prediction, and supply chain configuration. It highlighted visibility and collaboration as the most emphasised antecedents of SCRes, showing AI’s strength in improving transparency, adaptability, and proactive decision-making. Moreover, the study found that while AI supports the readiness, response, and recovery phases of SCRes, no significant work has yet addressed the growth phase, which is the stage where supply chains improve beyond their pre-disruption state. Notably, the AI-SCRes framework mapped AI techniques to SCRes antecedents and resilience phases. It also identifies gaps and proposes future research directions, particularly the integration of multiple AI techniques and a stronger focus on strategic resilience. In conclusion, the authors were able to consolidate separate bodies of work and offer practical insights for both academia and industry to harness AI in achieving robust and adaptive supply chains.

Building on this, Zhao et al. (2025) had analysed 126 peer-reviewed studies to examine how ML has been applied in carbon emission assessment. The study’s objective was to consolidate advancements in carbon accounting by evaluating the predictive accuracy, optimisation strategies, and driver factor selection capabilities of various ML models. Using data sourced from Web of Science between 2020 and 2024, the authors used keyword co-occurrence analysis and clustered the research into key themes. The findings showed that tree-based models such as RF and XGBoost were most commonly used due to their robustness and interpretability, followed by NN and SVMs. Critical application domains include energy consumption, urban planning, and construction, with a special emphasis on the importance of economic and demographic variables in predicting carbon emissions. Nonetheless, model limitations were also highlighted in the study. Overfitting, lack of data standardisation, and inconsistent performance evaluation metrics calls for the development of more interpretable, hybrid, and interdisciplinary ML frameworks. Overall, the authors have provided a robust foundation for integrating ML into carbon emission for forecasting and sustainability assessment, stressing its critical role in addressing climate change through improved decision-making and policy planning.

Similarly, a research titled *“Sustainability with Limited Data: A Novel Predictive Analytics Approach for Forecasting CO2 Emissions”* turned its attention to developing a predictive analytics method for forecasting carbon emissions in the aviation sector, even with limited or sparse data. The study responds to real-world disruptions such as the COVID-19 pandemic and geopolitical events, which have introduced irregularities and missing data in aviation emissions datasets. Utilising a proprietary dataset from the company Forecasting CO2 Emissions company (FCOE), consisting of monthly carbon dioxide (CO_2) emission records from 29,707 aircrafts over 32 months, starting from December 2018 to July 2021, the authors proposed a Non-Negative Adaptive Auto-Regression with Thresholding (NNAART) model based on Recursive Psuedoinverse Matrices and a dynamic "forgetting factor", denoted as $\boldsymbol{\lambda}$. From the study, it was found that the NNAART model outperformed traditional ML methods like Long Short-Term Memory NN (LSTM), Support Vector Regression (SVR), Generalised Regression Neural Network (GRNN), and ensemble techniques in both forecasting accuracy and computational efficiency. Plus, it was better suited for time series with a high number of zero or missing values. Unlike black-box AI methods, the proposed model was interpretable and allowed for real-time, constraint-sensitive forecasting. At the end of the study, the authors had demonstrated the deployment of their model in a commercial emissions forecasting platform, facilitating the data-driven decision-making process of stakeholders whilst complying with regulatory frameworks, such as the EU ETS and Carbon Offset and Reduction Scheme for International Aviation (CORSIA) (Filelis-Papadopoulos et al., 2024).

2.1.3 Multi-Criteria Decision-Making (MCDM) in Sustainable Supplier and Commodity Evaluation

Tronnebati et al. (2024) addressed the challenges in green supplier selection by proposing a MCDM framework in a study titled *"Green Supplier Selection Using Fuzzy AHP, Fuzzy TOPSIS, and Fuzzy WASPAS: A Case Study of the Moroccan Automotive Industry"*. The authors developed a framework that integrates Fuzzy Analytical Hierarchy Process (AHP) with Fuzzy Weighted Aggregated Sum-Product Assessment (WASPAS) and Fuzzy Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) to assess green supplier selection. A unique component of this research was its relevance to MCDM in sustainable supplier evaluation, as well as its comparative analysis of two hybrid MCDM models, that were, Fuzzy AHP-Fuzzy TOPSIS and Fuzzy AHP-Fuzzy WASPAS, which was uncommon in the existing literature. The dataset for this study was derived from a practical case study within the Moroccan automotive industry, involving four green suppliers evaluated by three decision-makers with extensive experience in supply chain management, quality management systems, and logistics. Fuzzy AHP was used to calculate the weights of criteria, whereas Fuzzy TOPSIS and Fuzzy WASPAS were applied to rank and select the best possible suppliers. The criteria for evaluation included five environmental factors, namely, "Health and security", "Sustainable product design", "Green packaging", "Investment recovery", and "ISO 14001 certification", alongside three additional conventional factors: cost, quality, and distribution. These factors were identified through literature review and expert input. According to the findings, both Fuzzy hybrid methods yield consistent rankings for green suppliers, with Supplier 3 consistently achieving the best ranking, thereby indicating the robustness and performance of the chosen models. In conclusion, these approaches have facilitated the process of selecting green suppliers under a Fuzzy environment for efficient ecological supplier selection in the global supply chain.

According to a research conducted by Baral et al. (2025), the authors focused on enhancing crude oil forecasting through Multi-Criteria Group Decision-Making (MCGDM). They aimed to describe and extend the Type-1 Fuzzy TOPSIS technique, specifically for handling uncertain data within MCGDM framework related to crude oil price forecasting. While the dataset directly involves a simulated crude oil price scenario for evaluating response choices, the core methodology centers on the application of the Fuzzy TOPSIS technique. This includes its extension for group decision-making, where multiple decision-makers assess various options against multiple criteria under a fuzzy environment. A notable contribution of this research, pertinent to the discussion of MCDM in commodity evaluation, was the detailed description of how Type-1 Fuzzy TOPSIS can be adapted for group decision-making to rate response choices in complex, uncertain scenarios, such as those impacting crude oil prices. This has showcased the versatility of Fuzzy TOPSIS in accommodating uncertainties and group consensus, which are critical elements in evaluating and ranking commodities and suppliers based on sustainability and risk-related criteria, although the specific application here was crude oil. The results of the study have demonstrated the effectiveness of the Fuzzy TOPSIS technique in providing a structured approach to decision-making under uncertainty, offering a valuable methodological insight for broader MCDM applications beyond crude oil, including sustainable supplier and commodity evaluation.

A related study by Khulud et al. (2023) comprehensively examined the advancements in sustainable supplier selection utilising MCDM approaches between 2013 to 2022. The dataset used for this bibliometric analysis consisted of 121 scientific publications sourced from the Scopus database. Keywords like “sustainable supplier selection” and “MCDM” were the main focus of the authors when conducting the analysis. Primarily, bibliometric analysis, coupled with graphical visualisation using the VOSviewer application to map research networks, collaboration patterns, and assess scientific impact. Additionally, the authors discovered that research regarding sustainable supplier selection through MCDM significantly grew, with a notable surge in publications in the year 2019. The authors also highlighted a gradual increase of research which combined methodologies like Fuzzy AHP (FAHP), TOPSIS, and analytic network process (ANP), plus Customer Relationship Management (CRM) for sustainable supplier selection. Overall, this study has underscored the increasing academic interest and importance of MCDM in addressing sustainability and risk criteria in supplier evaluation.

Further evidence was provided by Ardra & Barua (2025) whose goal was to develop a framework for examining and selecting sustainable suppliers for fresh fruits and vegetables within a circular supply chain, and to prioritise the most suitable supplier from available alternatives. In this research, the authors employed a dataset which was a case study involving a prominent supermarket chain in India that deals with fresh fruits and vegetables, with eleven decision factors contributing expert judgements. The methodology applied was a three-phase approach. The first phase was an extensive literature review and expert consultation to identify supplier selection indicators. Subsequently, the Best-Worst Method (BWM) was utilised to determine the weights of the selected criteria, followed by Fuzzy TOPSIS to prioritise and select the optimal supplier. Notably, this study also focused on circular supplier selection within the perishable food supply chain in a developing economy context, which had not received substantial scholarly attention previously. Ultimately, the results indicated that resource optimisation and circular competencies (ROC) and circular economy practices (CEP) emerged as the most crucial leading criteria for sustainable supplier selection. In short, the authors has demonstrated the effectiveness of the proposed framework which addresses the critical gap for selecting the right suppliers for fresher commodities in India.

Furthermore, an article titled *"A survey of multi-criteria decision making techniques for green logistics and low-carbon transportation systems"* focused on systematically reviewing over 190 papers published between 2010 and 2022 to clarify current practices and future directions of MCDM techniques in green logistics and low-carbon transportation systems. The dataset used for this review consisted of 195 refined papers, primarily journal articles and conference proceedings, sourced from the Web of Science database using keywords, such as "green", "MCDM", "logistics", "sustainability", and "transport". Here, the study employed a systematic literature review which involved material collection, descriptive analysis, category selection, and categorical evaluation of the collected studies which covered 14 classes of MCDM techniques. The findings revealed a significant upward trend in research in this area, particularly since 2015, with "logistics", "green", and "transport" brewing the most frequently used keywords in titles. Commonly applied MCDM methods include VIKOR, AHP, ANP, PROMETHEE, and various hybrid approaches, with a growing emphasis on handling uncertainty using novel fuzzy languages. All in all, this survey has integrated and reviewed MCDM applications across all key aspects of low-carbon transport and green logistics, filling a gap in existing holistic literature. (Tian et al., 2023)

2.1.4 Synthetic Data and Data Scarcity in Supply Chain Analytics

Based on a study by (Chatterjee & Byun, 2023) the authors addressed the issue of data scarcity in forecasting electric vehicle demand by proposing a novel synthetic data generation technique to improve prediction accuracy for electric kickboards, crucial elements in smart transportation networks. The present study utilised a dataset comprising real-time traffic volume, weather data, and holiday information from a specific city, which was then used as the basis for synthetic data generation. The primary method employed was a Conditional Generative Adversarial Network (CGAN)-based synthetic data generation technique. With the generated data from CGAN, various ML models were trained, including Random Forest Regressor (RFR), XGBoost, LightGBM, and CatBoost for demand prediction. The outcomes after the analysis has demonstrated that training these models on the synthetically generated data significantly enhanced their prediction accuracy across all models, with the RFR model exhibiting the most notable improvement. Thus, the study concluded that applying a CGAN-based approach to mitigate data scarcity in electric vehicle demand forecasting was effective, specifically for electric kickboards. This has also showcased a practical application of generative models for synthetic tabular data to fill supply chain data gaps and augment data for decision systems where historical information was limited.

In a complementary research, Hong et al. (2025) aimed to enhance predictive analytics in blood donation forecasting by leveraging GenAI to overcome data scarcity and variability issues, which is a critical challenge in supply chain management. The study's dataset was derived from real-world blood donation patterns, which served as the foundation for generating synthetic data. The methods primarily involved employing Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) to create synthetic datasets that mirrored these real-world donation patterns. In the next phase of the analysis, various ML models were trained on both real and synthetically augmented data, including RF, Linear Regression, NNs, and SVMs to evaluate their predictive performance. As a result, models trained on synthetically augmented data consistently outperformed those trained solely on real data in terms of predictive accuracy. In fact, models that were trained on GAN-generated data generally yielded more accurate forecasts than VAE-generated data. In summary, GenAI models have the ability to create high-quality synthetic tabular data, making them effective in addressing severe data gaps in a highly sensitive and critical supply chain like blood donation management, thereby enhancing risk modelling and decision systems where historical or complete data was often inaccessible.

Meanwhile, (Miletic & Sariyar, 2024) intended to evaluate the performance and trade-offs of various synthetic data generation methods, particularly in the context of augmenting limited datasets and addressing privacy concerns for sensitive information, which has been deemed highly relevant to handling missing or inaccessible supplier information in supply chain analytics. In the present work, diverse real-world medical datasets of varying complexity with both numerical and categorical attributes were used to test the efficacy of the synthetic data generation techniques. Several generative models, including GAN variants, such as CTGAN and CopulaGAN, a TVAE, and Copula were evaluated. The findings highlighted that while CTGAN was sensitive to training parameters, TVAE demonstrated robustness across different datasets, even for high-dimensional data, despite the incurred higher privacy risks. The authors have conducted a detailed comparative analysis of these specific generative models for tabular microdata, emphasising the persistent scalability challenges with GANs due to their computational resource demands. In conclusion, the research underscored that no single model universally excelled. The authors also stressed the need to understand the trade-offs between data utility and privacy, and to leverage the strengths of different models to effectively enhance synthetic data generation for applications where supply chain data gaps or inaccessible supplier information necessitated alternative data sources.

In a separate study titled *"Leveraging synthetic data to tackle ML challenges in supply chains: challenges, methods, applications, and research opportunities"*, the authors' objective was to systematically review the challenges of ML operations (MLOps) in supply chain tasks, specifically focusing on data privacy and scarcity, plus to explain how synthetic data could address these issues. 245 papers published between 2013 and 2024 were collected through a systematic literature review and used as the dataset for this investigation. The techniques involved classifying and analysing various synthetic data generation techniques, including generative models such as GANs, VAEs, and approaches using Large Language Models (LLMs) to assess their suitability for different supply chain problems. Results obtained from the study showed that data privacy and scarcity indeed represented significant barriers to the deployment and scaling of ML solutions in supply chains, and that synthetic data offered a promising avenue for data augmentation in risk modelling, enhancing decision systems, and mitigating supply chain data gaps, particularly concerning sensitive or inaccessible supplier information. Notably, the authors provided a holistic and structured framework that bridged synthetic data generation techniques with specific supply chain problems, proposing a comprehensive research agenda to further explore the potential of synthetic data in this domain (Long et al., 2025).

According to another similar study by Jackson et al. (2024), the authors were set out to propose a novel capability-based framework for analysing and implementing GenAI in Supply Chain and Operations Management (SCOM), thereby exploring its transformative potential. The study did not utilise a traditional empirical dataset as previously mentioned studies, rather, its foundation was a synthesis of existing literature and knowledge concerning GenAI and SCOM applications. The study at hand involved conceptual analysis and framework development, where key GenAI capabilities, namely perception, prediction, adaptation, and reasoning were identified and mapped to various SCOM functions. The findings indicated that GenAI could significantly enhance SCOM by improving decision-making, optimising processes, and revolutionising traditional approaches, particularly by addressing challenges related to data processing, forecasting, and risk management. Additionally, the findings also revealed the capability of GenAI and models that could produce synthetic data could augment scarce or inaccessible supply chain information, facilitate data augmentation in risk modelling, and bridge existing data gaps to enhance predictive analytics and complex decision systems in supply chain and operations management.

2.1.5 Clustering and Feature Selection Techniques in Risk Profiling

In a recent study by authors (N. Han & Um, 2024) they have investigated the intricate relationships between supply chain sustainability risks, global uncertainty, and various mitigating strategies necessary to achieve robust supply chain resilience capability. The dataset used for this study was collected through a survey, which provided the empirical basis for the analysis. Then, the authors employed a few methods, including SEM and moderation tests to empirically examine how sustainability risks and global uncertainty influenced supply chain resilience, as well as to identify the most effective risk management strategies. After a thorough analysis, the results suggested a structured procedure for enhancing supply chain resilience when confronted with diverse sustainability risks. Moreover, it provided insights into the optimal application of fundamental risk-mitigating strategies, namely accept, avoid, control, and share or transfer, depending on the specific sustainability risk environment. Notably, there was a lack of prior empirical research in this domain, as mentioned by the authors. Overall, this study highlighted complex risk modelling and decision systems in supply chains where data scarcity, particularly concerning comprehensive and historical risk-related information, could pose a significant challenge in the supply chain if neglected.

Sutrisno & Kumar (2023) proposed a novel model for assessing supply chain sustainability risk by integrating both subjective and objective perspectives of decision-makers, thereby addressing a research gap in current methodologies. The study made use of a case example from a SME producing handicrafts, which served as its dataset. A new decision support model that combined Failure Mode and Effect Analysis (FMEA) parameters with the Preference Selection Index (PSI) methodology for subjective weighting, and the Shannon entropy method for objective weighting of supply chain risk reprioritisation criteria was a unique technique applied in this study. At the end of the analysis, the results revealed critical sustainability risk dimensions and their associated risk elements that required management attention to foster more sustainable business operations. A unique discovery was that this was the first paper, to the authors' knowledge, to apply the integrated PSI and Shannon entropy method for evaluating supply chain risk based on five sustainability pillars, simultaneously considering the subjectivity and objectivity of decision-makers. In conclusion, this study has demonstrated a method for robust risk assessment in situations where data might be inherently subjective or incomplete, providing a framework for decision systems that could potentially be augmented by or derive insights from synthetic data when real-world information, such as precise supplier performance or risk data was scarce or inaccessible.

Moreover, the authors of the research paper *"Cluster-based supplier segmentation: a sustainable data-driven approach"* developed a novel clustering-based method for sustainable supplier segmentation, which supposedly was able to provide companies with suitable policies to control each segment more efficiently. This study comprised information from a real case study involving all suppliers of a company, considering both supplier characteristics and aspects of purchased items.

Closely like the methods employed by Ardra & Barua (2024), BWM was used to determine the weights of sub-criteria, followed by the application of the K-means clustering algorithm to segment the suppliers based on four identified criteria. The findings proved that utilising clustering algorithms enhanced the efficiency of supplier segmentation and facilitated the selection of optimal criteria for sustainable supplier segmentation and supplier relationship management. Additionally, the authors integrated sustainability considerations into the supplier segmentation problem through a data-driven clustering approach. Not only has this study highlighted the practical application of analytical methods for handling and organising existing supplier data, it has also stressed the importance of robust data-driven approaches in situations where supplier information might be incomplete or inaccessible, which implicitly suggests areas where synthetic data could be used for data augmentation in supplier risk modelling and decision systems by creating more comprehensive supplier profiles for better segmentation (Rahiminia et al., 2023).

According to a study by (Kang & Bhawna, 2025), they have explored the application of supervised ML classification models to address the complex issue of supplier performance analysis and risk profiling as a multi-class classification problem. They have also highlighted that this was a significant gap in the existing literature. Critically, the study used a synthetic dataset comprising 1,600 historical data points across 40 different suppliers, categorised into four distinct classes, directly addressing the common challenge of data scarcity in supply chain analytics. The techniques involved a structured approach to supplier selection and risk profiling using various supervised ML multi-class classification models, including Bagging Classifiers, alongside the analysis of prediction probabilities. The findings demonstrated that supervised ML models could be effectively applied to multi-class supplier performance and risk profiling problems, and the research identified specific models and their parameters that yielded optimal performance. Their innovative use of a comprehensive synthetic dataset to validate a multi-class classification approach for supplier risk profiling have provided a robust solution for data augmentation in risk modeling and decision systems. Nonetheless, the authors have also shown how such generative models could overcome the practical limitations of missing or inaccessible real-world supplier information in complex supply chain environments.

Jauhar et al. (2024) experimented on explainable AI (XAI) and attempted to optimise costs and profits while meeting customer demand in perishable supply chains by proposing a two-step solution framework. This framework involved segmenting customers based on their characteristics, then forecasting demand for each segment using transparent and responsible AI techniques. The dataset utilised for this study involved customer characteristics and associated demand data for perishable products. In this study, K-means and hierarchical clustering were employed for customer segmentation, along with XAI to model and compare various ML techniques for demand forecasting, specifically mentioning SVR and LSTM. Notably, the outcomes demonstrated that SVR outperformed autoregressive models and that the proposed two-step segmentation and forecasting process, particularly when using hierarchical clustering and LSTM, significantly surpassed conventional single-step unsegmented forecasting methods. In short, this study has showcased how leveraging granular customer characteristics could effectively mitigate data scarcity in predictive analytics for supply chains, allowing for more precise demand forecasting and improved decision-making even when historical sales data for specific segments was limited or inaccessible.

2.2 Discussion

The existing body of literature in GSCM highlights significant advancements in the integration of sustainability practices, particularly through DT, AI, and advanced analytical frameworks. Studies such as Singh et al. (2025) and Alabdali et al. (2025) have demonstrated that digital capabilities, including algorithmic management and absorptive capacity, are critical enablers of effective green supply chain performance. However, while these studies offer empirical support for the role of digital transformation and leadership in GSCM, they also reveal a common limitation, that is, the lack of integration between environmental performance metrics and data-driven decision-making models for risk mitigation. Most existing works focus on descriptive or diagnostic analyses of sustainability performance with fewer contributions addressing predictive or prescriptive capabilities in sustainability risk assessment.

Moreover, literature on AI and ML applications in SCRM underscores the growing relevance of predictive analytics in improving operational resilience. Works by Nezianya et al. (2024) and Gidiagba et al. (2025) illustrate that ML models such as RF, SVMs, and ANN outperform traditional approaches in risk identification and supplier evaluation. Nonetheless, several barriers persist, including limited sector-specific studies, integration challenges with legacy systems, and the interpretability of ML models. Another concern is the narrow scope of these studies, where environmental and sustainability risks are not always prioritised alongside financial and operational concerns, thus limiting their applicability for full-spectrum GSCM.

In the domain of MCDM, methodologies like Fuzzy AHP and Fuzzy TOPSIS have proven to be robust tools for supplier and commodity evaluation under uncertainty. For instance, Tronnebati et al. (2024) effectively combined these techniques with Fuzzy WASPAS to rank green suppliers based on sustainability criteria. However, there remains a gap in scalability and adaptability when these models are deployed in diverse industrial contexts. Additionally, while MCDM frameworks can offer structured evaluations, their dependency on expert judgments often introduces subjectivity and limits the automation potential for large-scale applications. Moreover, studies like (Ardra & Barua, 2025) stress the importance of integrating circular economy principles into GSCM models, which is an area that is still underrepresented in the current literature.

Furthermore, synthetic data generation has emerged as a promising solution to address the common problem of data scarcity in supply chain analytics. The application of GANs, TVAEs, and CTGANs, as seen in the works of Hong et al. (2025) and Miletic & Sariyar (2024), demonstrates the ability of generative models to enhance predictive accuracy and data quality. However, significant concerns remain regarding the balance between data utility and privacy, computational complexity, and model generalisability across domains. This has implications for supplier risk modelling, especially in environments where access to granular supplier data is limited or sensitive. Therefore, while the potential of synthetic data in risk analytics is promising, real-world validation and standardisation remain critical research gaps.

Finally, the literature on clustering and feature selection techniques for risk profiling, such as in the studies by (Rahiminia et al., 2023) and (Kang & Bhawna, 2025) provides evidence that data-driven segmentation using methods like K-means, DBSCAN, and XAI can significantly enhance supplier and commodity classification. These techniques support better decision-making by enabling more precise identification of risk levels and performance outliers. However, the effectiveness of these approaches is highly contingent on the quality of the input data and the appropriateness of the chosen features. While SHapley Additive exPlanations (SHAP) and other interpretable feature selection techniques help in explaining model outputs, their computational cost and dependency on specific ML models can hinder real-time application in dynamic supply chain environments.

In summary, the literature reviewed across the five thematic areas reveals a clear trajectory towards integrating intelligent systems, sustainability metrics, and risk analytics in supply chain management. Despite notable advancements, persistent challenges remain in terms of data accessibility, model scalability, real-time implementation, and the seamless integration of environmental criteria into mainstream risk evaluation systems. These limitations underscore the need for a consolidated, AI-augmented, and sustainability-focused risk management framework in GSCM, where it can bridge the methodological and technological gaps identified in the current state of research.

2.3 Conclusion

The review of existing literature reveals a growing convergence between sustainability objectives and data-driven technologies in the advancement of GSCM. Across multiple domains, from DT and ML, to MCDM and synthetic data generation, researchers have increasingly acknowledged the importance of integrating intelligent systems for enhanced supply chain sustainability and risk mitigation. Nonetheless, while the literature reflects meaningful progress, it also exposes several limitations and gaps that warrant further exploration. Many studies remain siloed in their focus, either emphasising operational efficiency, supplier evaluation, or sustainability metrics, but rarely integrating these components into a unified risk management framework tailored for GSCM.

Notably, predictive analytics using AI and ML has shown strong potential in improving supply chain resilience and sustainability assessment. Despite this, the adoption of such models is hindered by limited interpretability, lack of environmental focus in risk classification, and the unavailability of high-quality supplier data. Similarly, while MCDM methods such as Fuzzy AHP and Fuzzy TOPSIS have been extensively applied to rank suppliers under uncertainty, they often rely on subjective expert judgments and do not scale well across dynamic or large-scale datasets.

In addition to that, the incorporation of synthetic data, particularly through generative models like CTGAN and TVAE, offers a promising solution to the pervasive issue of data scarcity in supply chain analytics. This has opened new opportunities to augment ML applications and enable fairer, more representative risk assessments. Meanwhile, clustering and feature selection techniques, especially when combined with interpretable ML methods have emerged as valuable tools in profiling supply chain entities by sustainability risk levels. However, the lack of real-time adaptability and validation across varied sectors remains a constraint.

In conclusion, while individual methodologies such as MCDM, ML, clustering, and synthetic data generation have shown promise, the literature lacks an integrated, scalable, and environmentally focused framework that can guide GSCM risk management holistically. These insights underline the necessity for a novel, multi-layered framework that combines predictive analytics, sustainability metrics, and intelligent decision-making tools which are capable of supporting risk identification, evaluation, and mitigation across both commodity and supplier dimensions in green supply chains.

2.4 Chapter Summary and Evaluation

This chapter reviews prior studies relevant to GSCM and organises them into key thematic areas, such as sustainability integration, AI and ML applications in risk management, MCDM, data scarcity and synthetic data generation, and risk profiling techniques. The discussion synthesises findings to highlight research gaps, particularly the limited use of predictive modelling for sustainability risks and the lack of transparency in supplier data. Overall, this chapter provides a strong theoretical basis for the study, directly informing the design of the proposed framework.

Chapter 3

Theoretical Background

3 Theoretical Background

Chapter 3 presents the theoretical foundations that will be used in the proposed GSCM risk management framework. It introduces key concepts in predictive modelling, feature selection, and clustering for risk identification and grouping. This chapter also examines decision-making frameworks and optimisation techniques relevant to sustainability-focused SCRM. Additionally, it discusses approaches for addressing data scarcity through generative modelling and outlines statistical methods for quantifying relationships between risk factors and decision variables.

3.1 ML (ML) Models for Risk Prediction

3.1.1 Random Forest (RF)

RF is a widely adopted ML algorithm which aggregates the results of multiple decision trees to yield a singular and more accurate outcome. First formally introduced by Leo Brieman and Adele Cutler in 2001, RF serves as a powerful, non-parametric tool for both classification and regression tasks. It is suitable for various types of predictor variables without requiring prior assumptions about their association with the response variable. RF builds upon earlier ensemble methods like bagging and the random subspace method (Breiman, 2001). The primary problem it addresses is the instability and potential for overfitting inherent in single decision trees when they grow too deep. Therefore, by creating an ensemble of trees, RF aims to improve prediction accuracy and provide more stable results.

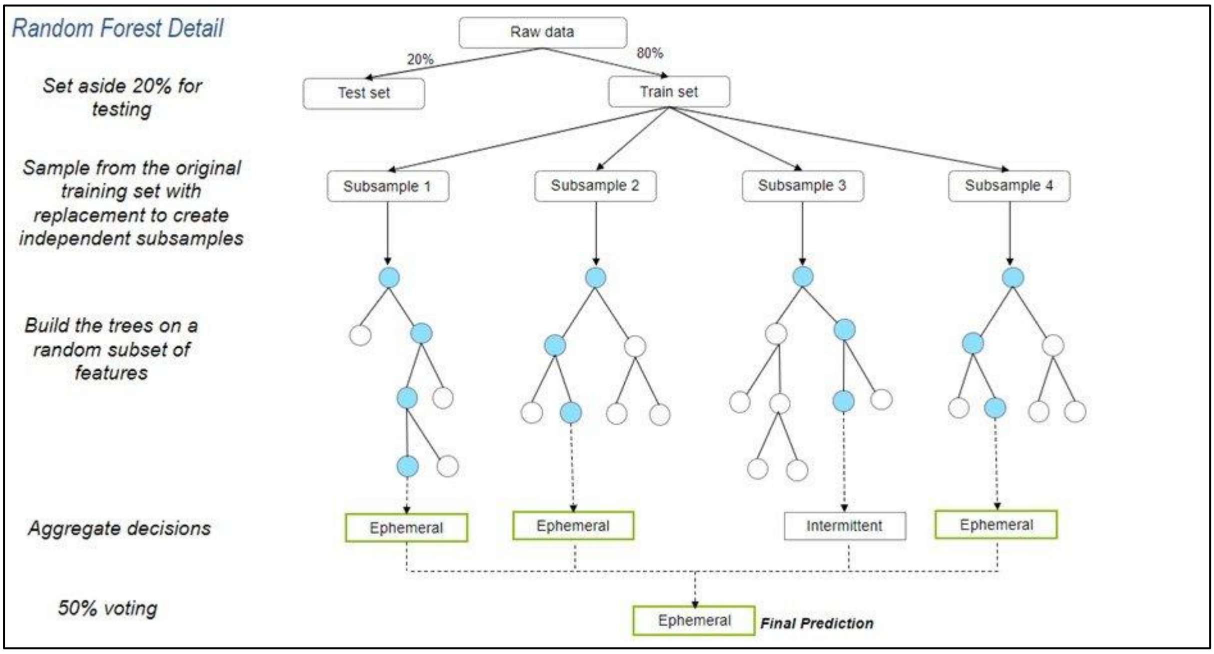

Figure 3.1.1: Overview of how RF works (United States Environmental Protection Agency, 2025).

RF operates by constructing a "forest" of uncorrelated decision trees through a combination of bagging and feature randomness. Each tree in the ensemble is independently grown using a bootstrap sample, which is a data sample drawn with replacement from the original training set. A portion of this training sample, typically one-third, is set aside as an out-of-bag (OOB) sample, which can be used for internal cross-validation and to estimate the generalisation error without the need for a separate test set. Furthermore, at each node within a tree, a random subset of features is selected, and the best split is determined from this subset rather than from all available features. This dual randomness in both data sampling and feature selection ensures that the individual trees are diverse and decorrelated, which is crucial for the algorithm's effectiveness. For regression problems, the final prediction is derived by averaging the predictions of all individual trees, while for classification, a majority vote among the trees determines the predicted class. Key hyperparameters such as the number of trees, the number of features sampled at each split (mtry), and the minimum node size influence the model's performance and require careful tuning (Biau, 2010; Probst et al., 2019).

Notably, RF significantly reduces the risk of overfitting, as the averaging of many decorrelated trees lowers the overall variance and improves generalisation performance. Its flexibility allows it to handle both regression and classification tasks with high accuracy, and it can effectively estimate missing values. Moreover, RF provides robust measures of feature importance, such as Gini importance or permutation importance, which can help in understanding the contribution of different variables to the prediction. Nonetheless, the algorithm can be computationally intensive and time-consuming when processing large datasets, due to the need to build numerous individual trees. It also demands more storage resources compared to single models because of the larger number of trees. Lastly, while individual decision trees are highly interpretable, the combined output of a vast number of trees makes the RF model less transparent and more complex to understand, often referred to as a “black box” model (IBM, 2025; Talekar, 2020).

3.1.2 Extreme Gradient Boosting (XGBoost)

In 2016, XGBoost was introduced by Tianqi Chen and Carlos Guestrin. At its core, XGBoost is a scalable end-to-end tree boosting system, which builds upon the concept of gradient tree boosting, also known as Gradient Boosting Machines or Gradient Boosted Regression Trees. The fundamental idea is to iteratively add new tree models that predict the residuals or errors of previous models, thereby progressively improving the overall prediction. Unlike traditional tree boosting, XGBoost incorporates a regularised learning objective that penalises the complexity of the model, helping to prevent overfitting. This regularisation term smooths the final learned weights, leading to more robust models. The model is trained in an additive manner, where each new function or tree is added to minimise the objective function, often utilising a second-order approximation for optimisation. Additionally, XGBoost employs techniques such as shrinkage, which scales down the influence of each new tree, and feature subsampling, which randomly selects a subset of features for each tree, further guarding against overfitting and speeding up computations (T. Chen & Guestrin, 2016).

One of the significant innovations of XGBoost lies in its algorithmic optimisations for scalability and efficiency. It features a novel sparsity-aware algorithm that can effectively handle missing values, frequent zero entries, and artefacts of feature engineering like one-hot encoding. This algorithm learns the best direction for instances with missing values, making computations linear to the number of non-missing entries and considerably faster than naive implementations. Furthermore, XGBoost incorporates a theoretically justified weighted quantile sketch procedure for approximate tree learning, enabling efficient handling of instance weights and allowing the algorithm to find optimal split points even with large datasets that do not fit into memory. It also exploits cache access patterns, data compression, and sharding to build a highly scalable system that can process billions of examples with fewer resources than existing solutions, supporting both parallel and distributed computing (T. Chen & Guestrin, 2016; Kavlakoglu & Russi, 2025).

For example, assume a financial institution predicting loan default risk. Traditional models might struggle with large datasets containing many features and missing information. Conversely, XGBoost, with its sparsity-aware algorithm, can efficiently process such data by automatically handling missing credit scores or income details by learning a default direction for these cases. Its ability to iteratively refine predictions by building an ensemble of trees means it can capture complex relationships in the data more effectively than a single model, leading to more accurate risk assessments. Additionally, the scalability features enable the institution to train models on massive historical datasets quickly, allowing for faster model exploration and deployment of more robust and accurate predictive systems. This combination of accuracy, flexibility, and computational efficiency makes XGBoost a favoured choice for demanding ML tasks.

3.1.3 Artificial Neural Network (ANN)

ANNs are computational models inspired by the structure and function of the human brain. They were first introduced by Warren McCulloch and Walter Pitts in 1943 as a logical model of neural activity, with further foundational work by Frank Rosenblatt in 1985 through the development of the perceptron. However, the real breakthrough came in 1988 with the introduction of the backpropagation algorithm, a method for training multi-layer networks, attributed to researchers such as Paul Werbos, Geoffrey Hinton, David Rumelhart, and Ronald Williams. ANNs are widely employed in tasks requiring pattern recognition, non-linear mapping, and predictive modelling, particularly in complex domains such as supply chain analytics, healthcare, and finance (Feldman et al., 1988; McCulloch & Pitts, 1943; Rosenblatt, 1958).

The architecture of ANN consists of layers of interconnected artificial neurons, where each layer serves a distinct function in the learning process. These layers include an input layer that receives raw data, one or more hidden layers where computation occurs, and an output layer that produces the final result. Fundamentally, each neuron processes inputs by applying weights, adding up the weighted inputs, and passing the result through a non-linear activation function such as ReLU, sigmoid, or tanh. This is to introduce complexity and allow the network to capture non-linear patterns in the data. During the training phase, the network adjusts these weights using the backpropagation algorithm, which computes gradients of the loss function, such as MSE or cross-entropy with respect to each weight, and updates them using gradient descent (GD). This iterative process is repeated over several epochs to minimise prediction error and improve model accuracy. Finally, the output layer produces the predicted result, which could be a class label in classification tasks or a numerical value in regression tasks. The connection strength between neurons, known as weights, and optionally the biases, are the learnable parameters updated during training to improve the model's performance (Bailer-Jones et al., 2001; S.-H. Han et al., 2018).

The learning process in ANNs can be summarised into four key steps: forward propagation to compute predictions, loss computation to evaluate performance, backpropagation to calculate gradients, and weight updates to refine the model. Forward propagation follows the aforementioned steps, whereas backpropagation propagates the error calculated backwards through the network to update the weights. These steps enable the network to capture complex, non-linear relationships between inputs and outputs. For example, in supply chain risk modelling, an ANN could learn intricate dependencies between environmental factors, like carbon dioxide (CO_2) emissions or energy usage and supplier performance metrics. Furthermore, the architecture of ANNs can be extended to deep learning models with multiple hidden layers, allowing for hierarchical feature extraction, especially useful in image processing or natural language understanding (NLU).

In practice, an optimisation algorithm, such as Adaptive Moment Estimation (Adam) is applied to update the network's weights and biases efficiently during training. Adam combines the advantages of two other extensions of stochastic gradient descent, namely, Adaptive Gradient Algorithm (AdaGrad) and Root Mean Square Propagation (RMSProp) by maintaining an exponentially decaying average of both past gradients and past squared gradients, known as first moment and second moment. These moment estimates allow Adam to adapt the learning rate for each parameter individually, making it especially suitable for problems with sparse gradients or noisy objectives. During each training iteration, the algorithm uses the calculated gradients from backpropagation to update each weight by applying a learning rate that is scaled based on these moment estimates. This adaptive behaviour accelerates convergence and helps avoid issues such as vanishing or exploding gradients, which are common in deep networks. By dynamically adjusting the step size for each parameter, Adam ensures a more stable and faster training process compared to traditional gradient descent methods, making it a widely used choice in the optimisation of deep neural network like ANNs (Werbos, 1990).

3.2 Feature Selection Techniques

3.2.1 Shapley Additive Explanations (SHAP)

SHAP is a model-agonistic interpretability framework that provides consistent, locally accurate attributions for ML predictions. It is grounded in a cooperative game theory, where the features of a predictive model are conceptualised as "players" contributing to a collective payout, that is, the model's prediction. SHAP frames the problem of explanation as on of the fairly distributing this payout among the features, ensuring that the attributions reflect each feature's contribution in a principled and consistent manner (Lundberg & Lee, 2017).

At its core, SHAP belongs to the class of additive feature-attribution models, which represent the explanation g for a specific instance as a linear function of binary feature indicators. Equation 1 shows the explanation model of the additive feature attribution methods:

$$g(v') = \emptyset_0 + \sum_{i=1}^{M} \emptyset_i z_i' \tag{1}$$

where $z' \in \{0, 1\}^M$ denotes whether each of the M features are present (1) or absent (0), and $\emptyset_i$ denotes the contribution of the feature i. SHAP's key theoretical contribution lies in showing that there exists a unique solution for $\emptyset_i$ that satisfies a set of three axioms:

- Local accuracy : Guarantees that the attributions sum to the actual prediction.
- Missingness : Ensures that the features with no effect have zero contribution.
- Consistency : Preserves monocity so that if a features' contribution increases across all possible coalitions in a modified model, its attribution will not decrease.

The unique solution satisfying these axioms is the Shapley value (Shapley, 1953), defined as the average marginal contribution of a feature across all possible feature subsets. For M features, the Shapely value for feature i is given by Equation 2:

$$\emptyset_i = \sum_{S \subseteq F\{i\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} \left[f_{S\cup\{i\}}\left(x_{S\cup\{i\}}\right) - f_S(x_S)\right] \tag{2}$$

where $N = \{1, \dots, M\}$ and $v(S)$ denotes the model output when only subset S of features is known. In the context of predictive modelling, $v(S)$ is typically defined as the conditional expectation of the model output given that the features in the subset take their observed values, which accounts for feature dependencies by integrating out the remaining features. This definition allows SHAP to fairly attribute contributions even when features are correlated.

Nevertheless, computing the exact Shapley values require summing over all 2^M possible subsets, which is computationally expensive for the models with many features. Thus, to tackle this problem, Lundberg & Lee (2017) proposed KernelSHAP, which is a model-agnostic approximation that reformulates the computation as weighted linear regression problem with a specifically designed Shapley kernel. For tree-based models, the authors introduced TreeSHAP, an exact polynomial-time algorithm that leverages the structure of decision trees to compute Shapley values efficiently. Moreover, variants like DeepSHAP approximate Shapley values for neural networks by combining SHAP with backpropagation-based techniques like DeepLIFT, which SHAP interaction values decompose contributions into main effects and pairwise feature interaction, offering richer insight into complex models.

Despite its strong theoretical foundations, the authors highlighted that the use of SHAP requires careful consideration of the chosen baseline and the method of handling missing features, as these choices influences interpretability and potential casual claims. Furthermore, while SHAP guarantees local accuracy and fairness under the chosen model and data distribution, its attribution reflects corelation and not necessarily causation, which should be considered in risk-sensitive decision-making contexts.

3.2.2 Recursive Feature Elimination (RFE)

Recursive Feature Elimination (RFE) is a widely used wrapper-based feature selection technique designed to improve model performance and interpretability by iteratively removing the least important features. It was first introduced in a seminal paper titled *"Gene Selection for Cancer Classification using Support Vector Machines"* (Guyon et al., 2002), where it was applied to microarray gene expression data, which was a challenging problem characterised by thousands of features but very few samples. The authors demonstrated that systematically removing irrelevant or redundant features improved classification accuracy and produced more interpretable models. Since then, RFE has become a standard method in ML for handling high-dimensional data, particularly in biomedical, financial, and image classification applications.

Conceptually, RFE follows a backward selection strategy. The process begins by training a model on the full feature set and computing an importance score for each feature. The least important feature or group of features are then removed, and the model is retrained on the reduced feature set. This elimination-and-retraining cycle is repeated recursively until a predefined number of features remain or until model performance reaches a stopping criterion. A key advantage of RFE over simple filter methods is that the feature ranking is recomputed at each iteration, so RFE accounts for feature interdependencies.

The most well-known variant, SVM-RFE, uses a linear SVM as its estimator. In a linear SVM, the decision function is given by Equation 3:

$$D(x) = w \cdot b + x \tag{3}$$

where $w = \{w_1, w_2, \ldots\ w_p\}$ is the weight of the vector representing feature contributions, and b is the bias term. Then, SVM learns w by solving the optimisation problem denoted in Equation 4:

$$\sum_{i} (w_i + w_i^* + C \sum_{k}^{i} \xi_k) \ subject\ to\ y_k[(w^* - w \cdot x_k + b)] \geq 1 - \xi_k \tag{4}$$

where (x_i, y_i) are the training samples, C is a regularisation parameter, and ξ_i are slack variables allowing soft margin classification. After the training process, the absolute magnitude of each weight $|w_j|$ serves as a natural importance score. At each RFE iteration, the feature j with the smallest $|w_j|$ will be removed.

Subsequently, the model is retrained on the reduced feature set, yielding a new w, and the entire procedure repeats until the desired subset of feature is selected. Notably, this process is greedy and does not revisit previously discarded features, nonetheless proven effective for discovering compact, discriminative feature subsets in practice.

3.3 Clustering Techniques for Risk Grouping

3.3.1 K-Means Clustering

K-Means and its conceptual origins trace back to 1956, where it was first discussed as partitioning a set into groups. The algorithm was then developed independently in 1957 at Bell Labs, in the context of vector quantisation. Later on in 1965, K-Means was published in a similar approach as well, and the name “K-Means” was coined by (MacQueen, 1967), who generalised the method and popularised its use in statistical data analysis. Together, these contributions have laid the foundation for what is now known as the Llyod-Forgy K-Means algorithm, or simply K-Means algorithm (Bock, 2007).

Mathematically, the goal of K-Means is to find a set of clusters and its corresponding centroids that minimise the intra-cluster sum of squares, which can also be referred to as the inertia. Meanwhile, the objective function expresses the total squared distance between points and their assigned cluster centroid, in which the algorithm seeks to minimise this quantity over all possible assignments of points to clusters and choices of centroids.

Fundamentally, the K-Means algorithm executes iteratively through two main steps, that are, assignment and update. The process begins with an initialisation step, where k centroids are chosen at random or using a more sophisticated initialisation technique, such as K-Means++ which spreads centroids more evenly and improved convergence. In the assignment step, each point is assigned to the cluster whose centroid is nearest in Euclidean distance. Once the assignments are made, the update step recomputes the centroid of each cluster as the average of its assigned points. Both assignment and update steps are repeated until convergence, which occurs when the centroids no longer change significantly or when point assignments stabilise. Each iteration is guaranteed to reduce the objective function or leave it unchanged, thus ensuring convergence to a local minimum. Notably, the results can vary depending on the initial centroid k selection.

When given large datasets, K-Means is computationally efficient, conceptually simple, and has good scalability. Nevertheless, it assumes clusters are roughly spherical and equally sized. Furthermore, it is sensitive to outliers and requires the number of clusters k to be specified a priori. Therefore, techniques such as the elbow method or silhouette score are often used in practice to determine an appropriate value of k by executing the algorithm multiple times with different initialisations to avoid poor local minima, as well as preprocess data to broaden inter-cluster distance, intra-cluster compactness, and overall improve cluster quality.

3.3.2 Density-Based Spatial Clustering (DBSCAN)

DBSCAN is a clustering algorithm introduced by Sander et al. (1998). Their work addressed the limitations of partition-based clustering methods, such as K-Means, which struggle with arbitrarily shaped clusters and are sensitive to outliers. DBSCAN introduced a density-based paradigm, enabling the discovery of clusters of arbitrary shape while simultaneously identifying noise points. The algorithm requires only two parameters, that are, the neighbourhood radius or epsilon ε (Eps) and the minimum number of points (MinPts) required to form a dense region.

According to the authors, for a point p, its ε-neighbourhood is formally defined as Equation 3:

$$N_{\varepsilon}(p) = \{q \, \epsilon \, D \mid dist(p, q \ \leq \ \varepsilon)\} \tag{3}$$

where D is the dataset and $dist(p, q)$ is a distance metric, typically Euclidean distance. Based on DBSCAN's concept, point p is a core point if $|N_{\varepsilon}(p)| \geq MinPts$. Points are then categorised as either core points, border points or noise points. Border points are points that lie within the ε-neighbourhood or a core point but do not themselves satisfy the core condition, whereas noise points are neither core nor border points, rather just outliers. Additionally, relational concepts like directly density-reachable and density-reachable. For instance, a point q is directly density-reachable from p if $q \in N_{\varepsilon}(p)$ and p is a core point. Meanwhile, q is density-reachable from p if there exists a sequence of points such that each $p = p_0, p_1, \ldots, p_n = q$ such that p_{i+1} is directly density-reachable from p_i. Clusters are then defined as maximal sets of the points that are density-connected, meaning any two points p and q in the cluster are connected through a chain of density-reachable points.

In mathematical terms, the algorithm finds all connected components in a graph where vertices represent data points and edges connect points with distance ε. This makes DBSCAN robust to noise and highly effective at finding clusters of arbitrary shape. Moreover, DBSCAN does not require prior knowledge of the number of clusters k, which has been deemed a major advantage for exploratory data analysis. Nonetheless, DBSCAN's performance and output quality depend heavily on appropriate selection of ε and MinPts. If ε is too small, many points may be labelled as noise, and if too large, clusters may merge.

3.4 Fuzzy MCDM Framework

3.4.1 Analytical Hierarchy Process (AHP) and Fuzzy Analytical Hierarchy Process (FAHP)

In its essence, AHP is a theory of measurement introduced by Thomas L. Saaty. It is a structured technique for organising and analysing complex decisions, especially those involving multiple criteria and subjective judgements. AHP works by breaking down a decision problem into a hierarchy of criteria and alternatives, where the root is the overall goal, leading to the subsequent nodes at each level are intermediate criteria, and finally the leaf, which is the lowest level of alternatives. The core of AHP involves pairwise comparisons, where decision-makers compare elements at each level of the hierarchy with respect to a common criterion from the level above. These comparisons are made using a fundamental scale of absolute judgments, typically starting from 1 to 9, where 1 indicates equal importance and 9 indicates extreme importance.

Table 1: The fundamental scale of absolute judgements (T. L. Saaty, 2008).

Intensity of Importance	*Definition*	*Explanation*
1	Equal Importance	Two activities contribute equally to the objective
2	Weak or slight	
3	Moderate importance	Experience and judgement slightly favour one activity over another
4	Moderate plus	
5	Strong importance	Experience and judgement strongly favour one activity over another
6	Strong plus	
7	Very strong or demonstrated importance	An activity is favoured very strongly over another; its dominance demonstrated in practice
8	Very, very strong	
9	Extreme importance	The evidence favouring one activity over another is of the highest possible order of affirmation
Reciprocals of above	If activity *i* has one of the above non-zero numbers assigned to it when compared with activity *j*, then *j* has the reciprocal value when compared with *i*	A reasonable assumption
1.1–1.9	If the activities are very close	May be difficult to assign the best value but when compared with other contrasting activities the size of the small numbers would not be too noticeable, yet they can still indicate the relative importance of the activities.

First and foremost, a pairwise comparison matrix is constructed with $n \times n$ dimensions, where n is the number of alternatives. Equation 4 shows an example of how the matrix will look like initially. From these pairwise comparison matrices, a priority vector or weight vector is derived, as shown in Equation 5 This vector represents the relative importance of each element.

$$A = \begin{bmatrix} 1 & a_{12} & a_{13} & \cdots & a_{1n} \\ \frac{1}{a_{12}} & 1 & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{1}{a_{1n}} & \frac{1}{a_{2n}} & \cdots & 1 & \cdots \end{bmatrix} \tag{4}$$

$$w_i = \frac{1}{n}\sum_{j=1}^{n} \hat{a}_{ij}\,, \quad w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \tag{5}$$

where $\hat{a}_{ij}$ is the normalised value of the relative importance of alternative i to criterion j.

Next, the consistency of these judgements is also measured using a consistency index (CI) and consistency ratio (CR). Basically, CI and CR function as a validation check on the weights assigned to the decisions. For instance, assume three items, A, B, and C. If item A is preferred over item B by 3 times and item B is preferred over item C by 2 times, item A must be preferred over item C by 6 times. Meanwhile, CR is derived by comparing CI with the average random consistency index (RI), where RI is a pre-calculated value based on the size of the matrix. These priority vectors are then synthesised by multiplying them by the priority of their parent nodes and cumulated for all such nodes to obtain an overall ranking of alternatives. (R. W. Saaty, 1987; T. L. Saaty, 2008) The following equations, Equation 6 and Equation 7, are the calculations for CI and CR respectively.

$$CI = (\lambda_{max} - n)/(n - 1) \tag{6}$$

$$CR = CI/RI \tag{7}$$

On the other hand, FAHP integrates fuzzy set theory with the traditional AHP to handle the inherent imprecision and subjectivity often present in expert judgements. While AHP uses crisp numbers for pairwise comparisons, fuzzy AHP uses fuzzy numbers, like triangular fuzzy numbers (TFNs) or trapezoidal fuzzy numbers to represent these judgements. A TFN is defined by three values: a lower bound (l), a most probable value (m), and an upper bound (u), denoted as (l, m, u). This allows decision-makers to express their preferences with a range rather than a single crisp value, better reflecting uncertainty. The process generally involves constructing fuzzy pairwise comparison matrices. Various algorithms exist for deriving fuzzy priority vectors from these matrices. A common approach is the extent analysis method, which involves calculating the fuzzy synthetic extent value for each criterion. For a set of n elements and m objectives, the fuzzy synthetic extent value for the i-th element is given by Equation 8.

Table 2: TFNs for the importance of criteria (Nădăban et al., 2016).

Rank	Attribute grade	Triangular FN
Very low	1	(0.00, 0.10, 0.30)
Low	2	(0.10, 0.30, 0.50)
Medium	3	(0.30, 0.50, 0.75)
High	4	(0.50, 0.75, 0.90)
Very high	5	(0.75, 0.90, 1.00)

$$S_i = \sum_{j=1}^{m} M_{ij} \otimes \left[\sum_{i=1}^{n} \sum_{j=1}^{m} M_{ij}\right]^{-1} \tag{8}$$

where M_{ij} are the fuzzy numbers representing the pairwise comparisons. After calculating the fuzzy synthetic extent values, a defuzzification process is often applied to convert fuzzy numbers into crisp values, and then a priority vector is normalised to obtain final weights. (R. W. Saaty, 1987; T. L. Saaty, 2008)

The primary difference between AHP and FAHP lies in how they handle human judgments and uncertainty. AHP uses crisp, exact numbers for pairwise comparisons, which can sometimes fail to capture the vagueness and imprecision inherent in human thought. In contrast, FAHP utilizes fuzzy numbers, allowing for a more nuanced and realistic representation of subjective preferences. The main benefit of FAHP over AHP is its ability to effectively deal with ambiguity and uncertainty. By using fuzzy numbers, FAHP can accommodate linguistic terms and imprecise judgments, making the decision-making process more robust and reflective of real-world complexities. This leads to more reliable and accurate priority weights, especially in situations where precise quantitative data is unavailable or difficult to obtain, and where expert opinions are crucial. FAHP also provides a more flexible framework for decision-makers to express their preferences, potentially leading to a higher degree of consistency in judgments by allowing for a range of values rather than forcing a single point estimate.

3.4.2 Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) and Fuzzy TOPSIS

TOPSIS was originally developed by (Chen Shu-Jenand Hwang, 1992) as a systematic method for comparing alternatives by identifying the solution that is closest to the ideal and farthest from the negative ideal, known as best solution and worst solution respectively. It assumes that the optimal solution should have the shortest geometric distance from the positive ideal solution (PIS) and the farthest from the negative ideal solution (NIS). However, classical TOPSIS uses precise numerical inputs, which may bot be sufficient in real-world decision-making scenarios, where human judgements are often vague or imprecise. Thus, fuzzy TOPSIS was introduced as an extension of TOPSIS to handle this uncertainty. By incorporating the fuzzy set theory, TOPSIS becomes particularly useful when linguistic or imprecise assessments are involved. (Nădăban et al., 2016)

TOPSIS evaluates each alternative against a set of criteria, normalises the data, applies criterion weights, and finally calculates each alternative's closeness to the ideal solution. This structured approach follows the steps below to rank alternatives in the order of their overall performance relative to the ideal outcome. (Zyoud & Fuchs-Hanusch, 2017) Firstly, the decision matrix (D) is constructed as per Equation 9.

$Step$ 1:

$$D = \left[x_{ij}\right] \tag{9}$$

where x_{ij} is the performance rating of alternative i with respect to criterion j.

Next, the decision matrix is normalised and represented as *R* in Equation 10.

$Step$ 2:

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x_{ij}^2}} \tag{10}$$

where *m* is the total number of alternatives, and the denominator is the Euclidean norm of the criterion of *j*'s column.

Moving on, the weighted normalised decision matrix (*V*) is calculated by obtaining the product between the weight of criterion *j* (w_j) and *R*, as shown in Equation 11.

$Step$ 3:

$$v_{ij} = w_j \cdot r_{ij} \tag{11}$$

The following step is to determine the PIS (A^+) and NIS (A^-).

$Step$ 4:

$$A^+ = \{\max(v_{ij}) \big| j \in J_{benefit}; min(v_{ij}) \big| j \in J_{cost}\} \tag{12.1}$$

$$A^- = \{\min(v_{ij}) \big| j \in J_{benefit}; max(v_{ij}) \big| j \in J_{cost}\} \tag{12.2}$$

Equation 12.1 shows the calculation for PIS, which consists of the maximum values for benefit or desirable criteria and minimum values for cost or undesirable criteria. On the other hand, Equation 12.2 shows the calculation for NIS, which consists of the minimum values for desirable criteria and the maximum values for the cost criteria. In general, PIS represents a hypothetical alternative that is best in all aspects, whereas NIS represents the worst possible scenario.

Then, the distances between A^+ and A^- are calculated. Using the Euclidean distance concept, this separation measure calculates how far each alternative is from the PIS and NIS. Other than quantifying the relative performance of each alternative, Step 5 helps identify which options are more aligned with the ideal outcomes, which ideally should have higher benefit and lower cost. Step 5 follows Equation 13.1 and Equation 13.2.

$Step$ 5:

$$S_i^+ = \sqrt{\sum_{j=1}^{n} \left(v_{ij} - A_j^+\right)^2} \tag{13.1}$$

$$S_i^- = \sqrt{\sum_{j=1}^{n} \left(v_{ij} - A_j^-\right)^2} \tag{13.2}$$

where A_j^+ and A_j^- are the ideal and non-ideal values for criterion j respectively. S_i^+ is the Euclidean distance from the PIS, whereas S_i^- is the Euclidean distance from the NIS.

Proceeding to Step 6, the relative closeness coefficient (CC) to the ideal solution is calculated with Equation 14.

$Step$ 6:

$$CC_i^* = \frac{S_i^-}{S_i^+ - S_i^-} \tag{14}$$

where CC_i^* is the relative closeness of alternative A_i to the ideal solution. Notably, this coefficient represents the degree to which an alternative resembles an ideal solution. If CC_i^* is close to 1, then the alternative is very close to the ideal solution and is highly preferred. Conversely, if CC_i^* is close to 0, then the alternative is very close to the worst-case scenario and is less desirable.

Finally, all the alternatives are consolidated and ranked in ascending order based on their relative CC_i^*, which represents how close each alternative is to the ideal solution and how far it is from the negative-ideal solution. A higher CC_i^* value indicates a more desirable alternative as it suggests greater proximity to the most favourable outcome and distance from the least favourable one. Therefore, the alternative with the highest CC_i^* is considered the optimal choice and will be selected as the best among the evaluated alternatives (Nădăban et al., 2016).

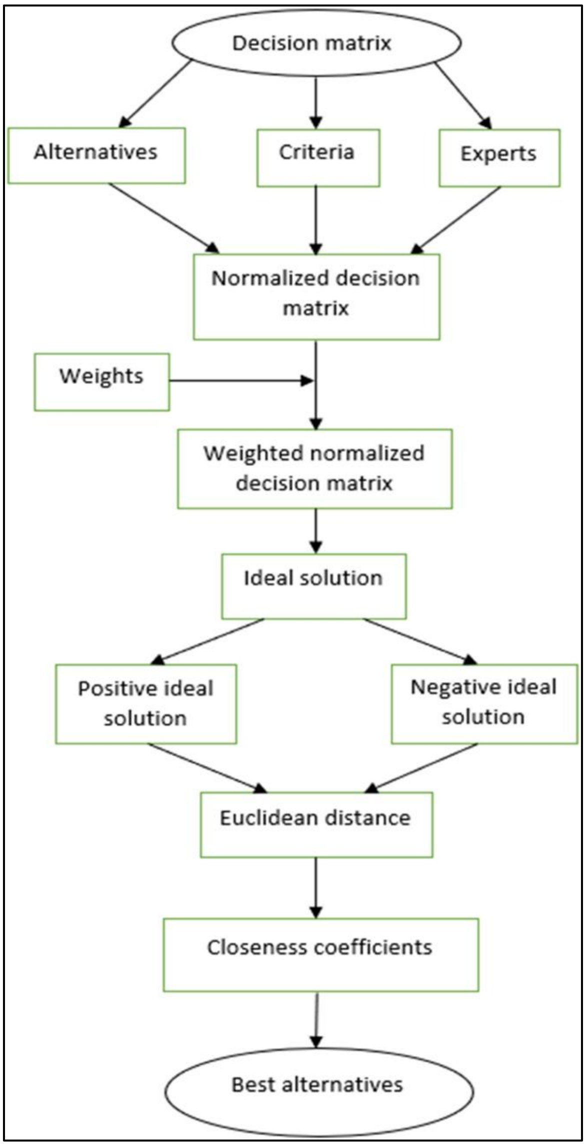

Figure 3.2.2: Overview of TOPSIS (Baral et al., 2025).

As previously mentioned, fuzzy TOPSIS is an integration of the classical TOPSIS method with the fuzzy set theory to handle the ambiguity and vagueness inherent in human judgements. While traditional TOPSIS crisp inputs, fuzzy TOPSIS allows decision-makers to express preferences using linguistic variables, such as "High", "Moderate", "Low", which are then represented as triangular fuzzy numbers (TFNs) of the form $\tilde{a} = (l, m, u)$.

Fuzzy TOPSIS begins by assigning a rating to the criteria and alternatives available. Then, the fuzzy ratings for alternatives $\tilde{x}_{ij} = (a_{ij}, b_{ij}, c_{ij})$ are aggregated across multiple fuzzy weights for criterion C_j, $\widetilde{w}_{ij} = (w_{j1}, w_{j2}, w_{j3})$, as shown in Equation 15 and Equation 16 respectively.

$$a_{ij} = \min_k\{a_{ij}^k\},\ b_{ij} = \frac{1}{K}\sum_{k=1}^{K} b_{ij}^k,\ c_{ij} = \max_k\{c_{ij}^k\} \tag{15}$$

$$w_{j1} = \min_k\{w_{j1}^k\}, w_{j2} = \frac{1}{K}\sum_{k=1}^{K} w_{j2}^k, w_{j3} = \max_k\{w_{j3}^k\} \tag{16}$$

Moving on, the normalised fuzzy decision matrix ($\tilde{R}$) is computed. The normalisation differs based on whether a criterion is a benefit (Equation 17.1) or a cost (Equation 17.2).

$$\tilde{r}_{ij} = \left(\frac{a_{ij}}{c_j^*}, \frac{b_{ij}}{c_j^*}, \frac{c_{ij}}{c_j^*}\right) \text{ and } c_j^* = \max_i\{c_{ij}\} \tag{17.1}$$

$$\tilde{r}_{ij} = \left(\frac{a_j^-}{c_{ij}}, \frac{a_j^-}{b_{ij}}, \frac{a_j^-}{a_{ij}}\right) \text{ and } c_j^- = \min_i\{a_{ij}\} \tag{17.2}$$

where a_{ij} is the lower bound or lowest possible value in fuzzy rating, b_{ij} is the most representative value, and c_{ij} is the upper bound or highest possible value in fuzzy rating. For instance, assume a TFN of $\tilde{x}_{23} = (5, 7, 9)$, which means for alternative 2 and criterion 3, the performance is at least 5, best estimate is 7, and at most is 9. Notably, a_{ij} represents the minimum assessment or optimistic view of the alternative *i* under criterion *j*, while c_{ij} represents the maximum assessment or pessimistic view of the alternative *i* under criterion *j*.

Similarly to TOPSIS, the weighted normalised fuzzy decision matrix ($\tilde{V}$) is computed using the normalised fuzzy ratings from the previous step in Equation 18.

$$v_{ij} = w_j \cdot \tilde{r}_{ij} \tag{18}$$

After $\tilde{V}$ has been computed, the fuzzy PIS (FPIS) and fuzzy NIS (FNIS) are calculated similarly as the TOPSIS method. Equation 19.1 and Equation 19.2 shows the calculation for FPIS and FNIS respectively.

$$A^* = (\tilde{v}_1^*, \tilde{v}_2^*, \ldots, \tilde{v}_n^*), \text{ where } \tilde{v}_j^* = \max_i\{\tilde{v}_{ij}^3\} \tag{19.1}$$

$$A^- = (\tilde{v}_1^-, \tilde{v}_2^-, \ldots, \tilde{v}_n^-), \text{ where } \tilde{v}_j^- = \min_i\{\tilde{v}_{ij}^1\} \tag{19.2}$$

The distance between each alternative to the FPIS and FNIS are then calculated. This step is shown in Equation 20.1 and Equation 20.2 respectively. Similarly, this step quantifies how "good" or "bad" an alternative is by identifying how close it is to the FPIS and how far it is from the FNIS.

$$d_i^* = \sum_{j=1}^{n} d(\tilde{v}_{ij}, \tilde{v}_j^*) \tag{20.1}$$

$$d_i^- = \sum_{j=1}^{n} d(\tilde{v}_{ij}, \tilde{v}_j^-) \tag{20.2}$$

Lastly, before ranking the alternatives, the CC for each alternative is determined by following Equation 21. The alternatives with higher values are preferred than those with lower values.

$$CC_i = \frac{d_i^-}{d_i^- + d_i^*} \tag{21}$$

In practice, the numerator uses the value for FNIS. If an alternative had a large d_i^- and small d_i^*, the CC would have a higher value, indicating that it is closer to the ideal solution and farther away from the non-ideal solution. However, the use of FPIS as the numerator is also accepted but not widely practiced, in theory. Otherwise, the CC will be interpreted inversely to the current interpretation, where a lower CC is desirable since it indicates the alternative to be closer to the non-ideal solution and farther from the ideal solution (Nădăban et al., 2016).

3.4.3 Fuzzy AHP-TOPSIS

Fuzzy AHP-TOPSIS is a hybrid MCDM methodology that integrates the Fuzzy AHP with the Fuzzy TOPSIS to address complex decision problems under uncertainty. The approach is designed to exploit the strengths of both Fuzzy AHP and Fuzzy TOPSIS. Fuzzy AHP systematically derives criteria weights under vagueness and subjectivity, while Fuzzy TOPSIS uses these weights to rank alternatives based on their closeness to an ideal solution. This integration is particularly suited to decision scenarios where criteria are both qualitative and quantitative, and where expert judgments may be imprecise or linguistically expressed.

The process begins by structuring the decision problem hierarchically and eliciting expert judgements of criteria importance through fuzzy pairwise comparisons. These are aggregated and processed to produce fuzzy criteria weights, which may be defuzzified into crisp values depending on the decision context. These weights are then used in the Fuzzy TOPSIS stage, where alternatives are assessed under each criterion, forming a weighted fuzzy decision matrix. The method identifies the FPIS, which is the best possible performance on all criteria, and the FNIS, which is the worst possible performance. Each alternative's distance to FPIS and FNIS is computed using an appropriate fuzzy distance measure, such as the vertex method or Euclidean distance for triangular fuzzy numbers. Finally, the closeness coefficient for each alternative is calculated as per Equation 18. Alternative are ranked in ascending order of CC_i, meaning the alternative closest to the ideal solution and farthest from the anti-ideal solution receives the highest ranking.

3.5 Optimisation Techniques for Fuzzy MCDM

3.5.1 Genetic Algorithm (GA)

GA are a class of evolutionary optimisation methods inspired by the principles of natural selection and genetics. The concept was formally introduced by Holland (1992) is his work *"Adaptation in Natural and Artificial Systems"*,where he proposed that solution search could mimic biological evolution through reproduction, crossover, mutation, and selection. A GA operates on a population of candidate solutions, each encoded as a chromosome which collectively evolve towards better solutions. The evolution is guided by a fitness function $f(x)$ that evaluates the quality of each solution x with the objective of either maximisation or minimisation. Then, the algorithm proceeds with the following steps:

1. An initial population is generated.
2. Fitness values are calculated.
3. Parent solutions are selected probabilistically based on fitness score, by roulette wheel or tournament selection.
4. Crossover recombines genetic material to create offspring.
5. Mutation introduces random variations to preserve diversity
6. Replacement determines which individuals form the next generation. This step often includes elitism to retain the best solutions.

Uniquely, the Schema Theorem proposed by Holland (1992) provides a theoretical foundation for the GA's ability to propagate and combine fit schemata over generations, gradually converging towards high-quality solutions.

3.6 Generative AI for Data Augmentation

3.6.1 Conditional Tabular GAN (CTGAN)

Authors Xu et al. (2019) introduced CTGAN in their paper *"Modeling Tabular Data Using Conditional GAN"*. The authors aimed to address specific challenges in generating realistic synthetic tabular data, especially when the data has both continuous and discrete types, imbalanced categorical variables, multi-modal continuous variables, and non-Gaussian distributions. According to the authors, traditional GANs had found success in image and text domain, but tabular data present unique difficulties. For example, discrete columns often have imbalanced categories, while continuous columns often show multiple modes. In addition to that, normalisation methods, such as simple min-max scaling may not preserve critical structure, but in turn produce vanishing gradient issues. Therefore, CTGAN was developed to better model such complexities.

Fundamentally, CTGAN uses a conditional generator that allows synthetic rows to be generated conditional on values of categorical variables, which ensures that the generated learned realistic joint distributions involving discrete features. Then, CTGAN employs a mode-specific normalisation for continuous columns. It identifies the modes of a continuous variable by fitting a mixture of Gaussian distributions and treat each mode separately. This facilitates better reproduction of multi-modal continuous feature shapes. Finally, it uses a training-by-sampling strategy to handle categorical imbalance, meaning rows are sampled in a way that ensures underrepresented discrete categories are adequately represented during training. This is to avoid situations where rare categories are not well modelled.

Furthermore, CTGAN is evaluated using a range of publicly available tabular datasets and simulated data to determine its ability to replicate both the statistical properties and practical utility of the original data. The distributional fidelity or likelihood fitness is measured to assess how closely the synthetic data preserved the original joint and marginal distributions, ensuring that generated samples were statistically indistinguishable from real data in terms of feature relationships, correlations, and category frequencies. Meanwhile, downstream ML utility was tested by training models such as logistic regression and decision trees on the synthetic data and evaluating their performance on real data. High performance in this stage indicated that the synthetic data successfully retained the decision-relevant patterns and predictive structures of the original dataset, making it a reliable substitute for real data in model development without causing significant performance loss.

3.6.2 Tabular Variational Encoder (TVAE)

Kingma & Welling (2013) introduced a scalable framework for learning in directed probabilistic models that involve continuous latent variables, where the posterior distributions are intractable. The authors proposed a variational inference method that uses a recognition network or encoder to approximate the posteriori, employing a reparameterisation trick. The reparameterisation allows stochastic gradient methods to efficiently optimise the variational lower bound, making inference practical for large datasets.

In the context of synthetic tabular data generation, VAEs provide a way to encode real data, both continuous and discrete into a latent space, then decode back to data space, capturing both the marginal distributions of features and their dependencies via the latent representation. The VAE framework ensures that latent variables are regularised to match a prior distribution, while simultaneously optimising reconstruction fidelity. Although TVAE modifies or adapts many details, like handling discrete features, imbalanced categories, ensuring joint dependencies, its core architecture relies on the framework of variational inference and encoder-decoder networks.

When evaluating synthetic data generation for tabular data, the principles from VAE theory, like inference, latent regularisation, reconstruction, encoder-decoder, are crucial because they help ensure that the generator does not simply memorise but learns generalisable representations. In TVAE, this implies that features with complex distributions are better preserved if the latent space is well structured, and the decoder is designed to handle mixed data types. The training methodology borrowed from VAE also affects how well the synthetic data preserves both global statistical properties, such as feature correlations and distributions, as well as downstream utility, which is how useful the generated data is for tasks like classification or regression tasks.

3.7 Statistical Modeling

3.7.1 Multiple Linear Regression (MLR)

MLR is a statistical method used to model the relationship between one continuous dependent variable and two or more independent variables. It extends simple linear regression in which only one predictor is involved. Since MLR involves several predictors, it enables modelling of more complex relationships in data.

In MLR, it is assumed that there is a relation in the population between the dependent variable Y and p independent variables $X_1, X_2, \ldots, X_p$. The general form of the population model is denoted as Equation 22:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon \tag{22}$$

where β_0 is the y-intercept, β_1 are the regression coefficients representing the change in Y associated with a one-unit change in X_i while holding other predictors constant. ε is the error term that reflects the deviation of actual observations from the prediction die to unobserved factors, measurement error, or intrinsic randomness.

From an observed data of n observations, the coefficients of a MLR, which are denoted as $b_0, b_1, \ldots, b_p$ are empirical estimated to approximate their population counterpart. Methods such as ordinary least squares (OLS) is determines the set of coefficients that minimises the sum of squared residuals, where a residual is the difference between an observed value and the value predicted by the regression model. In essence, OLS identifies the "line of best fit", which is the hyperplane in a p-dimensional predictor space that best explains the variability in the dependent variable across all observations. This approach ensures that the total prediction error is as small as possible, making OLS both computationally efficient and statistically optimal under the classical linear regression assumptions.

MLR relies on a set of well-defined assumptions that must hold to produce unbiased, efficient, and interpretable results.

1. Linearity	:	The dependent variable is a linear combination of the predictors and their coefficients
2. Independence	:	Observations and error terms are independent of each other.
3. Homoscedasticity	:	The error term has a constant variance.
4. Normality of errors	:	The error term is assumed to follow a normal distribution for hypothesis testing and confidence interval purposes.
5. No multicollinearity	:	Independent variables are not perfectly correlated. High multicollinearity inflates the variances of coefficient estimates, making them unstable and difficult to interpret, as small changes in the data can lead to disproportionately large changes in the estimated coefficients.

Each estimate regression coefficient b_i represents the unique effect of its corresponding predictor X_i on the dependent variable Y, holding all other predictors constant. This conditional interpretation is used to identify the individual contributions of multiple predictors in a complex, multivariate settings (Stanton, 2001).

3.8 Chapter Summary and Evaluation

Chapter 3 provides the theoretical support of the study, presenting the core principles behind predictive modelling, feature selection, clustering, decision-making frameworks, generative modelling, and statistical analysis relevant to GSCM risk management. The discussion is abstract enough to remain broadly applicable while still offering sufficient depth to justify the selected methods. All in all, this chapter effectively bridges the gap between the literature review and the methodological choices, ensuring conceptual clarity.

Chapter 4

Proposed Methodology

4 Proposed Methodology

This chapter outlines the methodology proposed for developing a comprehensive AI-driven risk management framework within the context of GSCM. The framework is designed to support the identification, assessment, and mitigation of sustainability-related risks at both the supplier and commodity levels, using an integrated combination of AI, ML, clustering techniques, synthetic data generation, and MCDM methods. Moreover, this chapter also outlines the experimental evaluation criteria employed to assess the performance and validity of the framework. These include a range of model-specific metrics used to evaluate the accuracy, consistency, and robustness of ML models, clustering results, and decision-making outputs. Each methodological stage is designed to be modular yet interdependent, contributing towards a reliable and scalable solution for GSCM.

4.1 Proposed Framework

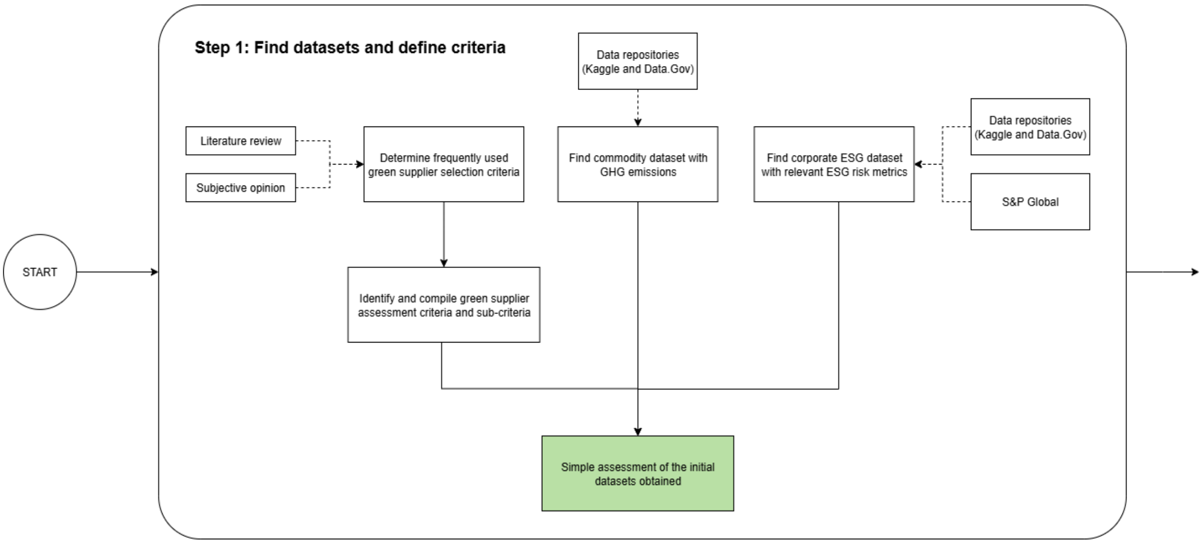

Figure 4.1(a): Phase 1 of the proposed framework.

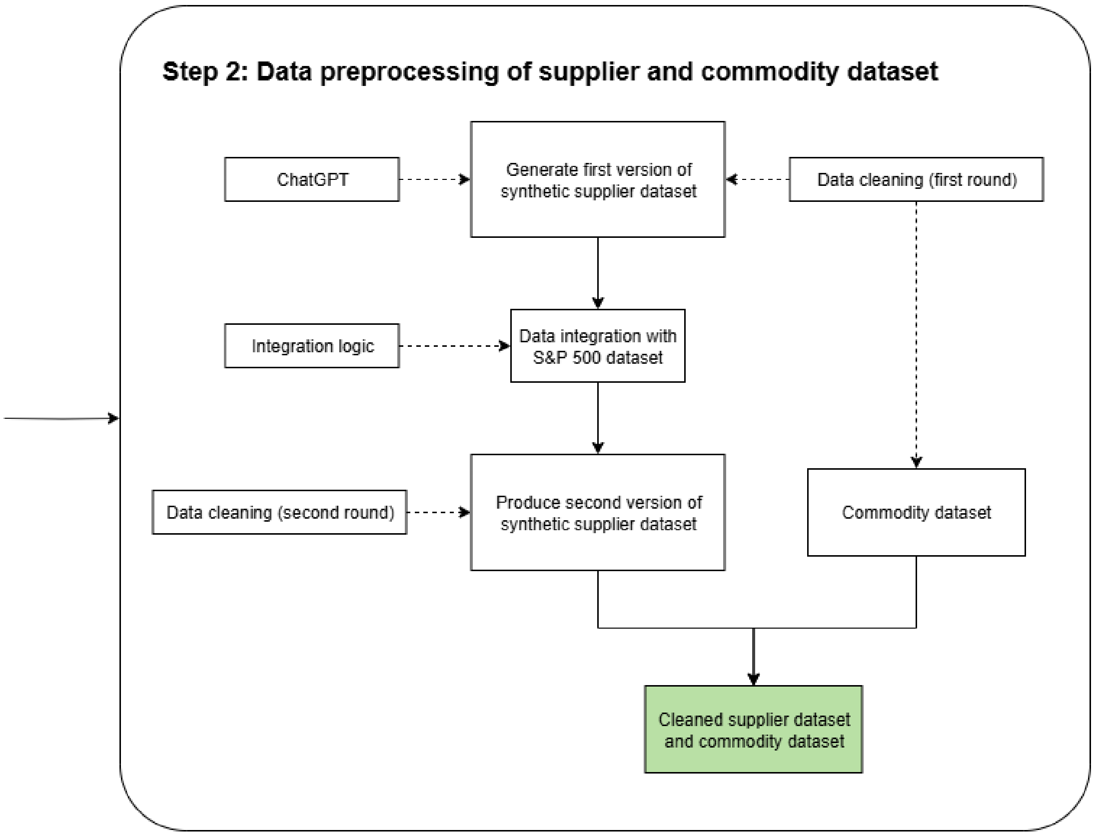

Figure 4.1(b): Phase 2 of the proposed framework.

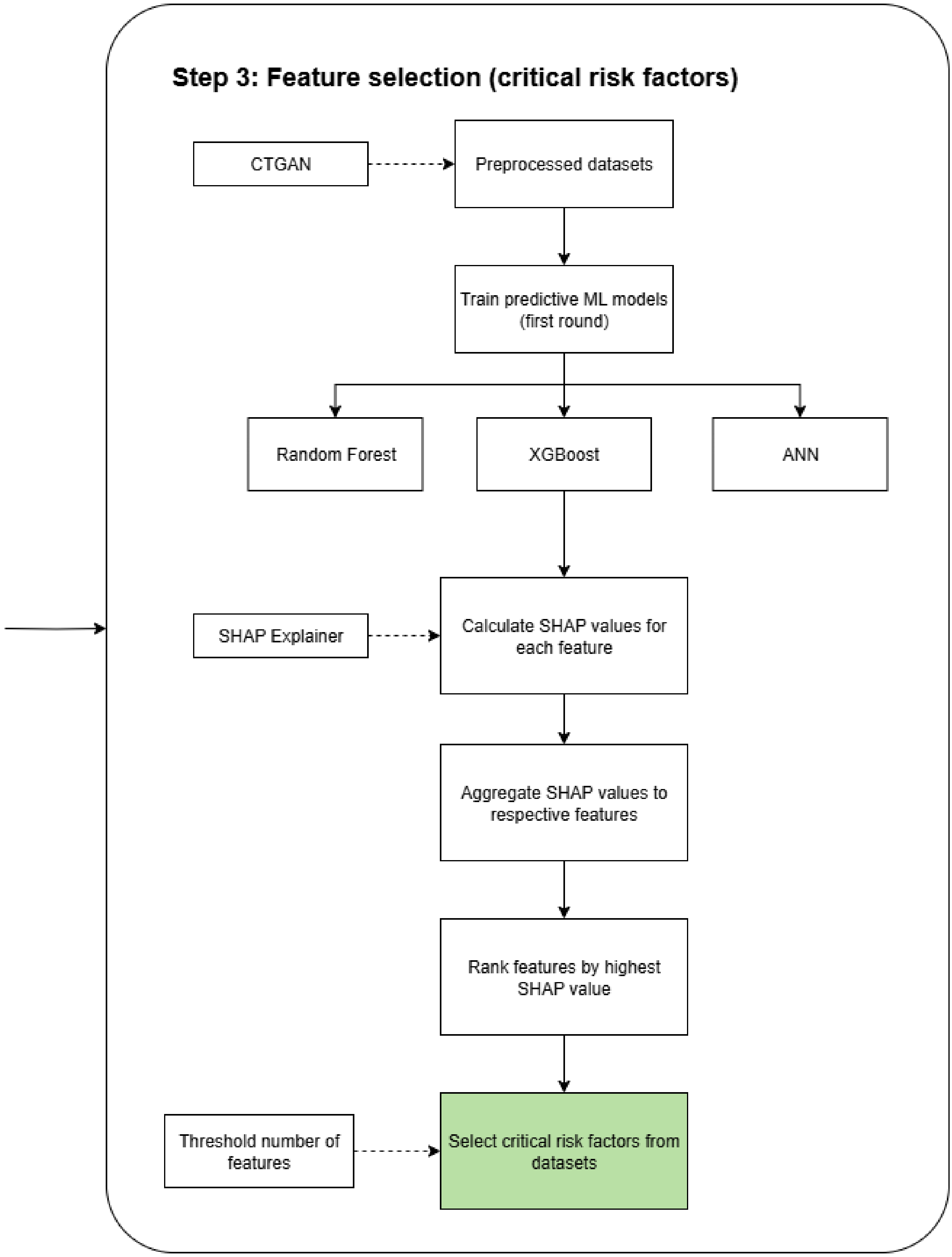

Figure 4.1(c): Phase 3 of the proposed framework.

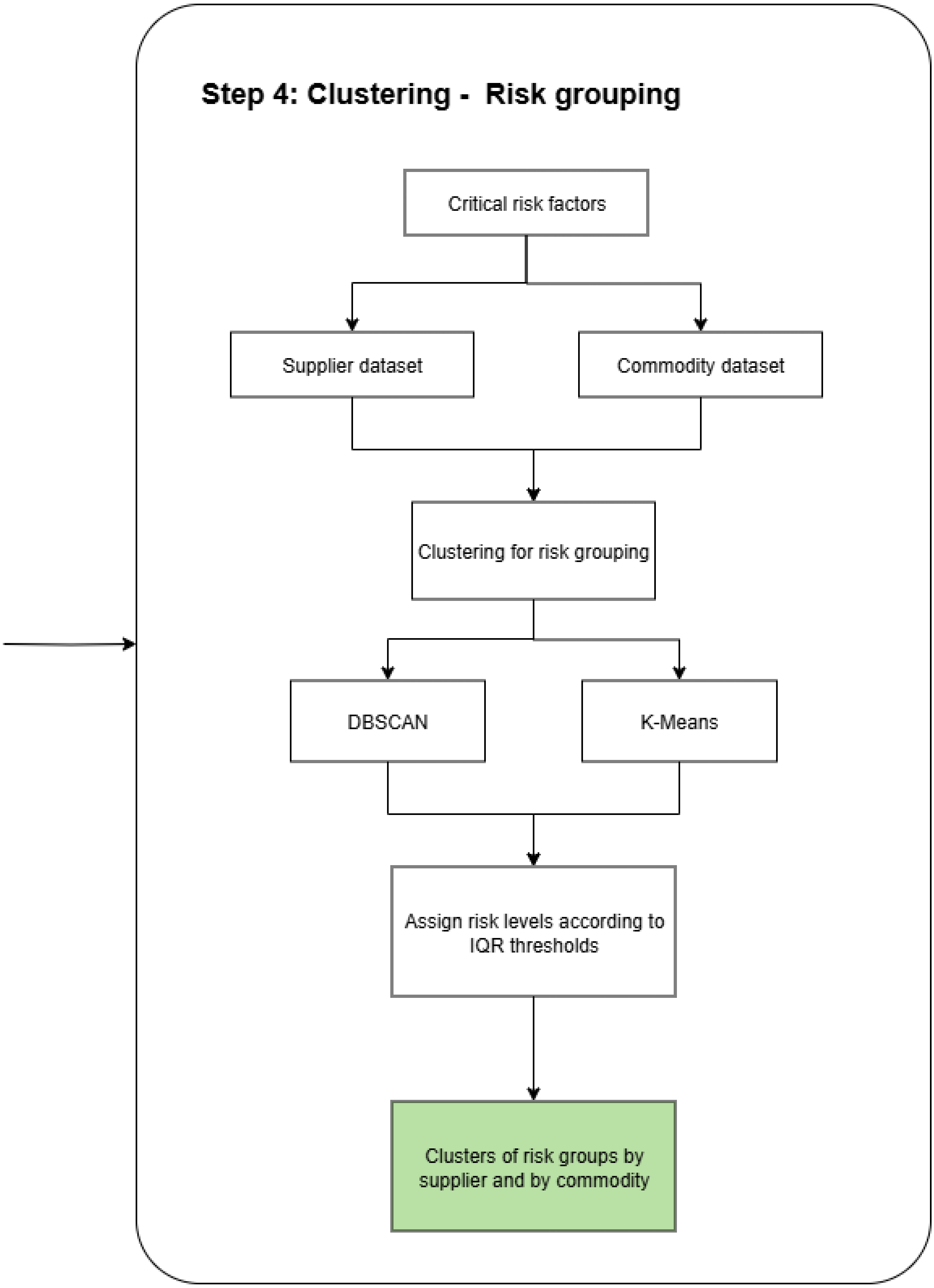

Figure 4.1(d): Phase 4 of the proposed framework.

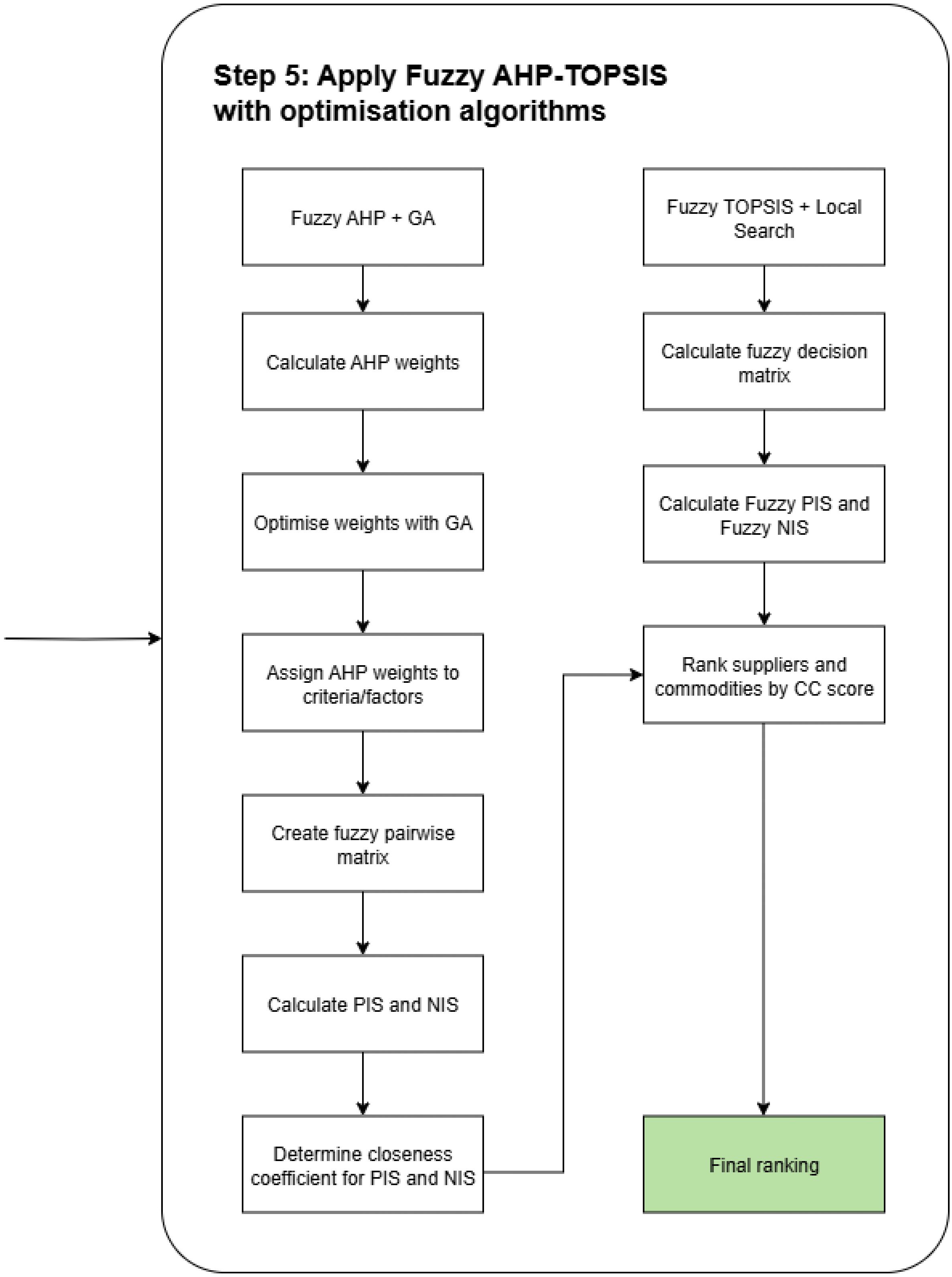

Figure 4.1(e): Phase 5 of the proposed framework.

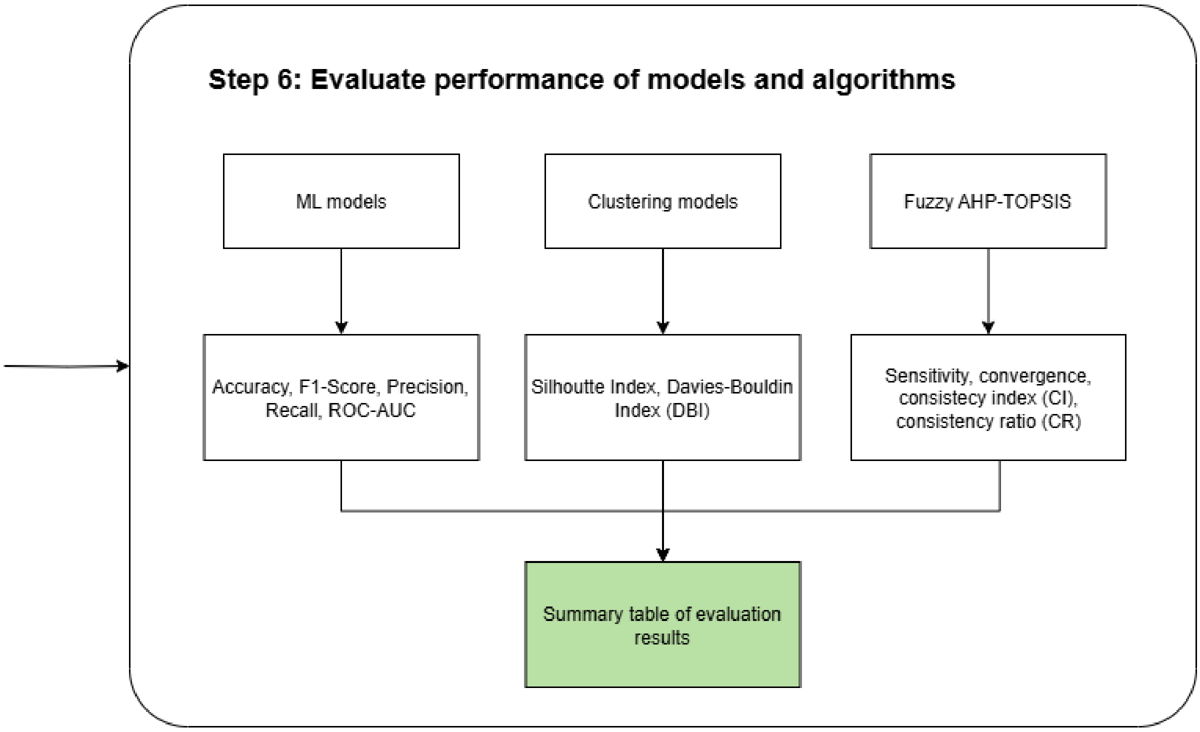

Figure 4.1(f): Phase 6 of the proposed framework.

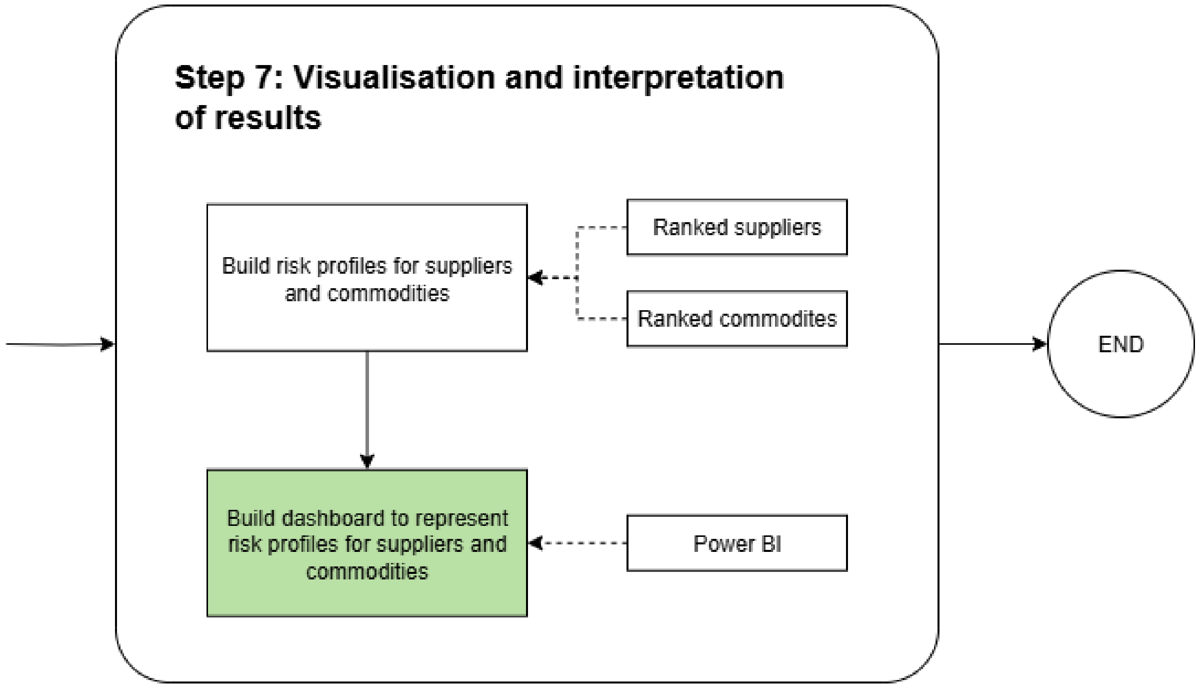

Figure 4.1(g): Phase 7 of the proposed framework.

Phase 1: Find datasets and define criteria

This initial phase involves the identification of relevant datasets and the establishment of suitable evaluation criteria for green supplier selection. The process begins with a literature review and subjective expert opinion to determine commonly used green supplier selection criteria. These sources help in identifying recurring themes, key factors, and metrics frequently adopted in academic and industry practices. Once these frequently used criteria are identified, they are compiled into a structured set of green supplier assessment criteria and sub-criteria.

In parallel, the search for appropriate datasets is conducted through reputable data repositories such as Kaggle and Data.Gov. The focus is on locating datasets that provide meaningful insights into commodity-related greenhouse gas (GHG) emissions and corporate-level environmental, social, and governance (ESG) risk metrics. The GHG emissions dataset serves as a proxy for the environmental impact of commodities, while the ESG dataset provides qualitative and quantitative risk indicators associated with suppliers or corporations. Data from providers like S&P Global is considered for its credibility and granularity.

Following the acquisition of these datasets, a simple yet critical assessment is performed to evaluate the quality, completeness, and relevance of the data obtained. This preliminary validation ensures that the data aligns with the green supplier selection criteria and that they are suitable for further processing and modelling in later stages.

Phase 2: Data preprocessing of supplier and commodity dataset

The second step focuses on the systematic preparation of the supplier and commodity datasets for downstream ML and analytical tasks. The process begins by generating an initial version of a synthetic supplier dataset using tools such as ChatGPT, which leverages structured logic and prompts to simulate a realistic dataset. This synthetic dataset acts as a proxy for real-world supplier data in the absence of publicly available, detailed supplier-specific information.

Once generated, the synthetic dataset undergoes the first round of data cleaning to ensure consistency, remove any anomalies, and standardise formats. In parallel, a commodity dataset which contains details such as emission factors is also cleaned to ensure compatibility. The cleaned synthetic dataset is then integrated with external datasets, including the S&P 500 dataset, using a predefined integration logic that aligns entities such as company names, industry sectors, or ESG scores.

After integration, a second version of the synthetic supplier dataset is produced, enriched with relevant ESG or financial indicators. This version is then subjected to a second round of data cleaning to eliminate redundant entries and to finalise the structure. At the end of this process, a well-curated and cleaned dataset, comprising both supplier and commodity-level information, is produced. This serves as the foundational input for modelling and risk analysis in the subsequent steps.

Phase 3: Feature selection (critical risk factors)

In the third step, the focus shifts to identifying the most critical risk factors within the preprocessed datasets through feature selection techniques. To enhance the robustness of the datasets, synthetic data generation using Conditional Tabular GANs (CTGAN) is applied. CTGAN is particularly suited for tabular data with complex interdependencies, helping to overcome data sparsity and improve the generalisability of models.

Following this, predictive ML models such as RF, XGBoost, and ANN are trained on the preprocessed datasets in the first modelling round. These models are selected based on their proven ability to handle structured data, learn feature interactions, and produce interpretable results when combined with explainability tools.

To determine the importance of each feature in driving predictions, Shapley values are calculated for each model. The SHAP Explainer provides local and global interpretations by assigning contribution scores to individual features. These SHAP values are then aggregated and analysed to rank features based on their overall impact on the model's output. This ranking enables the identification of the most influential risk factors.

Finally, based on a defined threshold for the number of features, a subset of high-impact features is selected. These critical risk factors form the basis for subsequent risk assessment, MCDM, and mitigation strategies within the green supply chain management framework.

Phase 4: Risk grouping with clustering

This step focuses on grouping suppliers and commodities into meaningful risk categories using clustering techniques. The process begins by taking the critical risk factors previously identified through feature selection and applying them separately to the supplier dataset and the commodity dataset. These datasets now contain only the most influential variables, making them suitable for unsupervised learning.

Clustering algorithms are then employed to uncover hidden patterns and group entities with similar risk profiles. Two prominent algorithms—DBSCAN and K-Means—are used in parallel to achieve this. This is because DBSCAN is advantageous for detecting clusters of varying densities and for handling noise in the data, while K-Means is efficient for forming compact, spherical clusters based on centroids.

Once clustering is complete, the next step involves assigning risk levels to each cluster. This is done using Interquartile Range (IQR) thresholds, which help in determining which clusters represent high, medium, or low risk groups. The outcome of this step is a set of labelled clusters that categorise suppliers and commodities according to their risk intensity, providing a structured basis for MCDM in the subsequent phase.

Phase 5: Apply Fuzzy AHP-TOPSIS with optimisation algorithms

In this phase, a hybrid decision-making framework combining Fuzzy AHP and Fuzzy TOPSIS (Fuzzy AHP-TOPSIS) is applied, enhanced with optimisation algorithms to improve accuracy and robustness. The process begins with Fuzzy AHP integrated with a Genetic Algorithm (GA). AHP weights are first calculated based on the green supplier selection criteria. These weights are then optimised using the GA, which helps refine the weight distribution to better reflect real-world decision-making priorities. The optimised weights are assigned to each criterion, followed by the construction of a fuzzy pairwise comparison matrix. This matrix is used to calculate the PIS and NIS, from which the closeness coefficients (CC) are derived to reflect each entity's proximity to the ideal state.

Simultaneously, Fuzzy TOPSIS enhanced with Local search (remove) is used to rank suppliers and commodities. This involves the formation of a fuzzy decision matrix, calculation of fuzzy PIS and NIS, and subsequent derivation of closeness scores. The Local search (remove) algorithm is employed to fine-tune these rankings by exploring neighbouring solutions to optimise the final closeness coefficients.

Ultimately, both approaches produce a comprehensive final ranking of suppliers and commodities, considering both subjective and objective factors, enhanced through algorithmic optimisation.

Phase 6: Evaluate performance of models and algorithms

This step is dedicated to evaluating the effectiveness and reliability of the models and methods employed throughout the project. The evaluation is categorised into three major domains: ML models, clustering models, and the Fuzzy AHP-TOPSIS decision-making framework.

For ML models, performance metrics such as accuracy, F1-ccore, precision, recall, and ROC-AUC are used to assess the predictive power and generalisability of the algorithms. These metrics provide a well-rounded view of how well the models can detect and classify sustainability risks.

Meanwhile, Clustering models are evaluated using Silhouette Index and Davies–Bouldin Index (DBI). These measures help determine the quality of the clusters formed, indicating whether the risk groups are well-separated and internally cohesive. A high silhouette score and a low DBI score are indicative of strong clustering performance.

Lastly, the evaluation of Fuzzy AHP-TOPSIS method involves checking the sensitivity, convergence, CI, and CR. These indicators ensure that the decision-making process is stable, logical, and resilient to changes in weights or inputs.

All evaluation results are compiled into a summary table, providing a comparative overview of each model and method's effectiveness. This summary informs stakeholders of the most reliable components within the overall framework.

Phase 7: Visualisation and interpretation of results

The final step focuses on interpreting the outcomes of the analysis and presenting them in a clear, actionable format. The project culminates in the construction of risk profiles for both suppliers and commodities, derived from the clustering and ranking processes. These profiles categorise entities based on their relative sustainability risks, enabling informed decision-making.

To facilitate stakeholder understanding and engagement, a dashboard is built using Microsoft Power BI to visualise the results. This dashboard dynamically displays ranked suppliers and commodities, their associated risk levels, and key metrics used in the evaluation process. It also supports interactive exploration, allowing users to filter and drill down into specific entities or risk factors.

The result is a comprehensive and user-friendly interface that transforms complex analytical outputs into strategic insights. This supports supply chain managers and sustainability officers in making transparent, data-informed decisions regarding supplier engagement and risk mitigation strategies.

4.1.1 Data Source and Collection

4.1.1.1 Supplier Dataset

As previously highlighted in the problem statement, publicly accessible supplier information remains limited due to confidentiality and data privacy restrictions. An extensive search across multiple online database repositories like Kaggle, Data.Gov, GitHub and UCI ML Repository have been conducted. However, no suitable supplier datasets were identified. Although a potential dataset was located on Data.Gov, it contained fewer than 20 records and features attributes that were largely irrelevant to the objectives of this study. Consequently, it was excluded from further consideration.

In response to this data scarcity, and as recommended by several prior studies (Fang et al., 2024; Rahiminia et al., 2023; Tronnebati et al., 2024; Varchandi et al., 2024; Zulqarnain et al., 2024), a comprehensive literature review was conducted to identify widely accepted supplier evaluation criteria. In addition to the general criteria suppliers should have, green and sustainable dimensions have also been incorporated for facilitating the construction of a synthetic supplier dataset. Table 3 presents the selected criteria used for assessing and ranking supplies based on sustainability performance.

Table 3: Commonly used sustainable supplier evaluation criteria in the literature.

Dimension	Supplier Selection Criteria	Sub-Criteria	Sources
General	Cost	Unit price	(Tronnebati et al., 2024; Varchandi et al., 2024)
		Total cost of ownership	
		Total acquisition cost	
	Quality and reliability	Defect rate	(Tronnebati et al., 2024; Varchandi et al., 2024; Zulqarnain et al., 2024)
		Performance consistency	
		Product durability	
		Customer satisfaction	
	Delivery	On-time delivery	(Tronnebati et al., 2024; Varchandi et al., 2024)
		Lead time	
	Service	After-sales support	(Rahiminia et al., 2023; Varchandi et al., 2024)
		Responsiveness	
	Compliance and certifications	ISO 9001 Certification	(Rahiminia et al., 2023; Zulqarnain et al., 2024)
		Regulatory compliance	
Environmental Sustainability	Sustainable product design	Recyclability of product or service	(Tronnebati et al., 2024)
		Energy efficiency of product or service design	
		Minimises or prevents use of harmful products in manufacturing process	
	Green packaging	Biodegradability of packaging material	(Tronnebati et al., 2024)
		Reusability of packaging	
		Minimise number of materials used for the packaging	
		Minimise the use of potentially hazardous components in the packaging	
	GHG emissions	CO_2 emissions	(Varchandi et al., 2024)
		GHG emission reduction targets	
	Energy consumption	Energy efficiency	(Rahiminia et al., 2023; Varchandi et al., 2024)
		Usage of renewable energy	
	Environmental-friendly material	Sustainable raw materials	(Varchandi et al., 2024)
		Recyclable components	
	Waste and e-waste management	Recycling rate	
		Waste reduction programs	

Dimension	Supplier Selection Criteria	Sub-Criteria	Sources
		E-waste reduction	(Rahiminia et al., 2023; Tronnebati et al., 2024; Varchandi et al., 2024)
	ISO 14001 Certification	Reduce negative environmental impacts on organisations	(Tronnebati et al., 2024; Zulqarnain et al., 2024)
		Reduce usage of resources and waste	
		Indirectly influence supply chain stakeholders to choose more environmentally sustainable activities	
		High ESG score	
Social Sustainability	Social management commitment	Corporate Social Responsibility (CSR) initiatives	(Varchandi et al., 2024)
	Worker's contract	Labour rights compliance	(Rahiminia et al., 2023; Tronnebati et al., 2024; Varchandi et al., 2024)
		Employee benefits	
	Employment insurance	Ensure long-term employment of workers	(Varchandi et al., 2024)
	Health and security	Closely monitor production safety	(Tronnebati et al., 2024)
		Protecting employees' health	
		Appropriate training on health and safety	
	Supplier willingness	Willingness for open and honest communication	(Rahiminia et al., 2023)
		Willingness for gender equality	
		Willingness for racial fairness	
		Willingness for transparent information sharing	
		Willingness for training programs on environmental issues	
Resiliency	Supply chain resilience	Business continuity	(Fang et al., 2024; Varchandi et al., 2024)
		Risk management	
	Backup supplier	Alternate sourcing solutions	(Rahiminia et al., 2023; Varchandi et al., 2024; Zulqarnain et al., 2024)
		Redundancy plans	
	Visibility	Supply chain transparency	(Varchandi et al., 2024)

Dimension	Supplier Selection Criteria	Sub-Criteria	Sources
		Real-time tracking	
	Capacity and scalability	Reserve capacity	(Varchandi et al., 2024; Zulqarnain et al., 2024)
		Surge capacity	
	Flexibility	Production flexibility	(Varchandi et al., 2024)
		Order customisation	
Economic	Profit impact	Environmental costs	(Rahiminia et al., 2023)
		Resource consumption	
	Cost efficiency	Procurement cost	(Rahiminia et al., 2023)
		Total cost	
	Investment recovery	Selling of surplus inventory or materials	(Tronnebati et al., 2024)
		Selling of scrap or waste materials	
		Selling of excess capital equipment	
Risk	Supply risk	Accessibility	(Rahiminia et al., 2023; Zulqarnain et al., 2024)
		Carbon disclosure	
		Number of available suppliers	
		Substitution possibilities	
		Environmental legal requirements	
Digitalisation	E-commerce	-	(Fang et al., 2024)
	Digital marketing	-	
	Customer experience	-	
	Data analysis	-	

4.1.1.2 Commodity Dataset

The commodity dataset utilised in this study, titled *"SupplyChainGHGEmissionFactors_v1.2_NAICS_byGHG_USD2021.csv"* (Arvidsson, 2023) was publicly accessible through both Data.Gov and Kaggle. This dataset was selected due to its relevance and comprehensiveness in capturing environmental emission data across a wide range of commodities and industry sectors. It has provided essential information required for assessing the sustainability and risk associated with commodity-level operations in GSCM.

The dataset contains 16,256 entries with eight attributes. It comprises of greenhouse gas (GHG) emission factors linked to various North American Industry Classification System (NAICS) commodity codes. Specifically, it presents the emission intensity in kilograms of carbon dioxide (CO_2)-equivalent (kg CO_2e) per United States dollar ($USD) of economic output, allowing emissions to be evaluated relative to production value. Additionally, the data also distinguishes between different types of GHGs, including CO_2, methane (CH_4), and nitrous oxide (N_2O), along with their corresponding economic sectors, such as manufacturing, agriculture, mining, and services. This structure enabled granular analysis of carbon intensities at the commodity level, which was essential for risk modelling and sustainability assessment. Furthermore, the dataset was formatted in alignment with 2021 economic valuation, using consistent dollar-year references to ensure comparability across commodities. It served as a reliable foundation for quantifying environmental risks associated with traded goods, particularly when integrated with supply chain decision variables later in the study. Table 4 shows a summary of the attributes in the dataset.

Table 4 Summary of attributes in the commodity dataset.

Attribute	Data Type	Description
2017 NAICS Code	Integer	A six-digit NAICS code that represents the specific industry or commodity sector.
2017 NAICS Code	String	The descriptive title of the NAICS industry or commodity.
GHG	String	The type of greenhouse gas used in emission calculation.
Unit	String	The measurement unit used.
Supply Chain Emission Factors without Margins	Float	The emission factor, excluding any profit margin considerations.
Margins of Supply Chain Emission Factors	Float	The range or uncertainty margin associated with the emission factor.

Attribute	Data Type	Description
Supply Chain Emission Factors with Margins	Float	The emission factor that includes margins to reflect realistic supply chain variation.
References USEEIO Code	String	Internal reference code used in the U.S. Environmentally-Extended Input-Output (USEEIO) model.

4.1.1.3 Other Related Datasets

S&P Global (2025) is a leading provider of financial information, analytics, and credit ratings, renowned for delivering data-driven insights to global markets. Headquartered in New York, the organisation operated through several divisions, including S&P Global Ratings, S&P Global Market Intelligence, and S&P Dow Jones Indices, which all serve as stakeholders in financial, corporate, government, and academic sectors. In the context of sustainability, S&P Global plays a critical role by offering robust Environmental, Social, and Governance (ESG) ratings and climate risk analytics. Enabling organisations to benchmark and improve their sustainability performance. Through tool such as the S&P Global ESG Scores and the Sustainable1 platform, the company helps investors and businesses make informed decisions that align with long-term environmental goals and regulatory frameworks. These ESG risk metrics provide an authoritative benchmark for modelling supplier sustainability and risk exposure in GSCM. By referencing S&P Global's data, this research ensures alignment with industry-recognised standards in sustainability risk assessment.

The *"SP 500 ESG Risk Ratings.csv"* dataset, which was found on Kaggle and compiled by (Dorobantu, 2023), contains information on 503 publicly listed companies. It provides a detailed breakdown of ESG risks based on each corporate listed. The dataset comprises of fifteen attributes, including company identifiers, such as ticker symbols and names, operational data, including sector, industry, and human capital, and ESG metrics, as presented in Table 5. Notably, the ESG-related attributes specifically include individual risk scores for ESD dimensions, a total ESG risk score, and a categorised risk level. Additionally, the dataset provides a controversy level and controversy score to indicate the extent of any reported ESG-related issues linked to the company. Ultimately, this dataset was chosen since it was deemed useful for modelling supplier-level ESG behaviour, which the present literature lacks.

Table 5: Summary of attributes in the S&P500 Titans' Ethical Odyssey dataset.

Attribute	Data Type	Description
Symbol	Object	The ticker symbol representing the company.
Name	Object	Full name of the company.

Attribute	Data Type	Description
Address	Object	The registered address of the company's headquarters.
Sector	Object	The economic sector in which the company operates.
Industry	Object	The specific industry classification of the company.
Full Time Employees	Object	Number of full-time employees.
Description	Object	General business description of the company.
Total ESG Risk Score	Float	Overall ESG risk score, where lower ESG risk score is better.
Environment Risk Score	Float	Sub-score indicating environmental-related risk.
Governance Risk Score	Float	Sub-score indicating governance-related risk.
Social Risk Score	Float	Sub-score indicating social-related risk.
Controversy Level	Object	Categorical indicator of ESG controversies, represented as either: • 'None Controversy Level' • 'Low Controversy Level' • 'Moderate Controversy Level' • 'High Controversy Level' • 'Significant Controversy Level' • 'Severe Controversy Level'
Controversy Score	Float	A numerical score that represents the severity of controversies.
ESG Risk Percentile	Object	The percentile ranking of the company's ESG risk among other companies.
ESG Risk Level	Object	A qualitative categorisation of the ESG risk, represented as either 'Low', 'Medium', or 'High', 'Severe', or 'Negligible'.

4.1.2 Experimental Evaluation Criteria

To assess the effectiveness and reliability of the proposed AI-driven risk management framework for Green Supply Chain Management (GSCM), a series of experimental evaluation criteria were established across multiple analytical components, including ML models, clustering algorithms, MCDM techniques, and generative models. Each component has played a distinct role in achieving the research objectives and was evaluated using appropriate quantitative and qualitative metrics to ensure the validity and robustness of the results.

For the ML models employed in risk factor prediction and classification, standard performance metrics such as accuracy, F1-score, precision, recall, and ROC-AUC are utilised. These indicators collectively provide insights into the predictive power, generalisability, and balance of the models, particularly in the context of imbalanced or complex sustainability datasets. The inclusion of both threshold-dependent and threshold-independent metrics allows for a more nuanced evaluation of classification performance.

The clustering algorithms, namely DBSCAN and K-Means, are evaluated based on internal validation measures such as the Silhouette Index and Davies–Bouldin Index (DBI). These metrics assess the cohesion within clusters and the separation between clusters, respectively, helping to determine the effectiveness of the grouping of suppliers and commodities according to sustainability risk profiles. A higher Silhouette score and a lower DBI value indicate stronger clustering performance.

For the Fuzzy AHP-TOPSIS framework, supported by optimisation techniques, evaluation focuses on the consistency and stability of the decision-making process. Metrics such as the CI and CR are applied to ensure logical coherence in the pairwise comparisons of criteria. Additionally, sensitivity analysis and convergence testing are conducted to validate the robustness of the final rankings against changes in weights or input variations.

In the case of synthetic data generation, particularly with the use of CTGAN, qualitative assessment is complemented by distributional comparison and statistical validation methods. These checks help confirm that the generated data preserve the underlying structure and variability of the original datasets without introducing bias or noise.

Together, these experimental evaluation criteria ensure that each component of the proposed framework performs reliably, contributes meaningfully to the research objectives, and supports transparent, data-driven decision-making in green SCRM.

4.2 Summary of Proposed Framework

This study proposes a comprehensive, AI-driven framework for risk identification, analysis, and decision-making in the context of GSCM with a focus on supplier and commodity sustainability risks.

The framework integrates data-driven ML techniques, synthetic data generation, clustering algorithms, and MCDM methods to support robust and explainable risk profiling for sustainable supplier evaluation.

The proposed work is structured into seven sequential steps. It begins with the identification of relevant datasets and the definition of green supplier selection criteria, derived from both literature review and expert judgement. Publicly available datasets on GHG emissions and ESG risk metrics are assessed alongside a compiled list of evaluation dimensions and sub-criteria. Synthetic supplier datasets are generated where gaps exist and are further integrated with credible sources such as the S&P 500 to enhance data richness and realism.

Subsequently, the data undergoes thorough preprocessing, including multi-stage cleaning and formatting, to ensure consistency and usability. The cleaned datasets are then used to train ML models such as RF, XGBoost, and ANN, which are applied to extract and rank critical risk factors. Then, feature importance is determined using SHAP values to ensure transparency and interpretability. These critical features are then utilised for unsupervised clustering, where DBSCAN and K-Means algorithms are employed to group suppliers and commodities into distinct risk clusters. Risk levels are assigned based on IQR thresholds, enabling categorisation into low-, medium-, and high-risk groups.

To enhance decision-making, the framework applies a hybrid Fuzzy AHP-TOPSIS approach, optimised with GA. Fuzzy AHP is used to assign weights to evaluation criteria through pairwise comparisons, while Fuzzy TOPSIS ranks suppliers and commodities based on closeness coefficients relative to ideal solutions. These steps ensure that subjective and objective factors are systematically incorporated into the final decision model.

Evaluation of the proposed models and algorithms is conducted using a combination of statistical and algorithmic performance metrics, including classification accuracy, ROC-AUC, clustering validity indices like Silhouette Index and DBI, and decision consistency measures like CI and CR. This is followed by the development of risk profiles and a Power BI dashboard that visualises ranked suppliers and commodities, offering a user-friendly interface for interpretation and strategic planning.

In summary, the proposed framework offers an integrated, scalable, and explainable solution for sustainability-focused risk management in supply chains. It addresses data limitations through synthetic augmentation, ensures interpretability through explainable AI, and supports structured decision-making via optimised MCDM methods, thereby contributing significantly to the advancement of GSCM practices.

4.3 Chapter Summary and Evaluation

This chapter describes the research design, dataset, tools, and techniques employed to implement the proposed framework. It outlines the development of predictive models, the application of decision-making frameworks, and the use of generative models for data augmentation. The methodology is systematic and reproducible, making it a strong foundation for achieving the research objectives.

References

Alabdali, M. A., Yaqub, M. Z., & Windsperger, J. (2025). Green digital leadership and algorithmic management for sustainable supply chains: A serial mediation model. *International Journal of Information Management Data Insights*, *5*(1), 100343. https://doi.org/10.1016/J.JJIMEI.2025.100343

Ardra, S., & Barua, M. K. (2025). Sustainable supplier selection among supermarket's fresh fruits and vegetable supply chains based on circular practices in India. *Environment, Development and Sustainability*, 1–37. https://doi.org/10.1007/S10668-024-05347-1/TABLES/3

Arvidsson, J. (2023, April 20). *Supply Chain Greenhouse Gas Emission Factors*. Kaggle. https://www.kaggle.com/datasets/joebeachcapital/supply-chain-greenhouse-gas-emission-factors

Asih, H. M., Sutrisno, A., Wuisang, C. E. A., & Faishal, M. (2025). Sustainability decision-making in poultry slaughterhouses: A comparative analysis of AHP and fuzzy AHP. *MethodsX*, *14*, 103193. https://doi.org/10.1016/j.mex.2025.103193

Atieh, A. A., & Abushaega, M. M. (2025). Achieving Supply Chain Sustainability Through Green Innovation: A Dynamic Capabilities-Based Approach in the Logistics Sector. *Sustainability 2025, Vol. 17, Page 5716*, *17*(13), 5716. https://doi.org/10.3390/SU17135716

Bailer-Jones, C. A. L., Gupta, R., & Singh, H. P. (2001). An introduction to artificial neural networks. *Hardware Architectures for Deep Learning*, 3–26. https://doi.org/10.1049/PBCS055E_ch1

Balkumar, K., Gedam, V. V., Himateja, M., Anbuudayasankar, S. P., Narassima, M. S., Ganesh, K., Dwarakanath, M., & Pazhani, S. (2024). Prevalence and future trend in green supply chain management (GSCM): a systematic literature review. *Benchmarking*, *31*(6), 2002–2033. https://doi.org/10.1108/BIJ-06-2022-0363/FULL/PDF

Baral, S. P., Parida, P. K., & Sahoo, D. (2025). Fuzzy TOPSIS technique for multi-criteria group decision-making: A study of crude oil price. *Results in Control and Optimization*, *19*, 100565. https://doi.org/10.1016/J.RICO.2025.100565

Biau, G. (2010). Analysis of a Random Forests Model. *Journal of Machine Learning Research*, *13*, 1063–1095. https://arxiv.org/pdf/1005.0208

Bock, H.-H. (2007). Clustering Methods: A History of k-Means Algorithms. In G. and B. P. and de C. F. Brito Paula and Cucumel (Ed.), *Selected Contributions in Data Analysis and Classification* (pp. 161–172). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-540-73560-1_15

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32. https://doi.org/10.1023/A:1010933404324/METRICS

Chatterjee, S., & Byun, Y. C. (2023). A Synthetic Data Generation Technique for Enhancement of Prediction Accuracy of Electric Vehicles Demand. *Sensors 2023, Vol. 23, Page 594*, *23*(2), 594. https://doi.org/10.3390/S23020594

Chen Shu-Jen and Hwang, C.-L. (1992). Fuzzy Multiple Attribute Decision Making Methods. In *Fuzzy Multiple Attribute Decision Making: Methods and Applications* (pp. 289–486). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-46768-4_5

Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. https://doi.org/10.1145/2939672.2939785

Chen, Y., & Guo, H. (2024). Corporate performance: green supply chain management, digital transformation and carbon neutrality. *Management Decision*. https://doi.org/10.1108/MD-12-2023-2344

Culot, G., Podrecca, M., & Nassimbeni, G. (2024). Artificial intelligence in supply chain management: A systematic literature review of empirical studies and research directions. *Computers in Industry*, *162*, 104132. https://doi.org/10.1016/J.COMPIND.2024.104132

Dhillon, M. K., Rafi-Ul-Shan, P. M., Amar, H., Sher, F., & Ahmed, S. (2022). Flexible Green Supply Chain Management in Emerging Economies: A Systematic Literature Review. *Global Journal of Flexible Systems Management 2022 24:1*, *24*(1), 1–28. https://doi.org/10.1007/S40171-022-00321-0

Dhumras, H., & Bajaj, R. K. (2024). On potential strategic framework for green supply chain management in the energy sector using q-rung picture fuzzy AHP & WASPAS decision-making model. *Expert Systems with Applications*, *237*(Part B), 121550. https://doi.org/10.1016/j.eswa.2023.121550

Dorobantu, D. M. (2023). *ESG Scores Showdown-S&P500 Titans' Ethical Odyssey*. Kaggle. https://www.kaggle.com/code/dianaddx/esg-scores-showdown-s-p500-titans-ethical-odyssey

Dwidienawati, D., Indrajaya, B. L., & Zanten, E. Van. (2025). Factors Affecting the Implementation of Green Supply Chain in Companies in Indonesia: A Qualitative Study. *Sustainability 2025, Vol. 17, Page 5349*, *17*(12), 5349. https://doi.org/10.3390/SU17125349

Fang, J., Zhou, W., & Xiong, L. (2024). Multi-criteria decision making approach for supplier selection and order allocation in a digital supply chain resilience. *Annals of Operations Research*. https://doi.org/10.1007/s10479-024-06435-1

Feldman, J. A., Fanty, M. A., & Goodard, N. H. (1988). Computing with structured neural networks. *Computer*, *21*(3), 91–103. https://doi.org/10.1109/2.34

Filelis-Papadopoulos, C. K., Kirshner, S. N., & O'Reilly, P. (2024). Sustainability with Limited Data: A Novel Predictive Analytics Approach for Forecasting CO2 Emissions. *Information Systems Frontiers*, 1–25. https://doi.org/10.1007/S10796-024-10516-8/FIGURES/13

Gidiagba, J. O., Tartibu, L. K., & Okwu, M. O. (2025). A systematic review of machine learning applications in sustainable supplier selection. *Decision Analytics Journal*, *14*, 100547. https://doi.org/10.1016/J.DAJOUR.2025.100547

Gupta, I., Raeisi, S. T., Correa, S., & Wicaksono, H. (2025). Evaluating risk factors in automotive supply chains: A hybrid fuzzy AHP-TOPSIS approach with extended PESTLE framework. *Journal of Open Innovation: Technology, Market, and Complexity*, *11*(1), 100489. https://doi.org/10.1016/j.joitmc.2025.100489

Guyon, I., Weston, J., Barnhill, S., & Vapnik, V. (2002). Gene Selection for Cancer Classification using Support Vector Machines. *Machine Learning*, *46*(1–3), 389–422. https://doi.org/10.1023/A:1012487302797

Han, N., & Um, J. (2024). Risk management strategy for supply chain sustainability and resilience capability. *Risk Management*, *26*(2), 1–26. https://doi.org/10.1057/S41283-023-00138-W/TABLES/5

Han, S.-H., Kim, K. W., Kim, S., & Youn, Y. C. (2018). Artificial Neural Network: Understanding the Basic Concepts without Mathematics. *Dementia and Neurocognitive Disorders*, *17*(3), 83. https://doi.org/10.12779/DND.2018.17.3.83

Hariyani, D., Hariyani, P., Mishra, S., & Sharma, M. K. (2024). A literature review on green supply chain management for sustainable sourcing and distribution. *Waste Management Bulletin*, *2*(4), 231–248. https://doi.org/10.1016/J.WMB.2024.11.009

Holland, J. H. (1992). Genetic Algorithms. *Scientific American*, *267*(1), 66–73. http://www.jstor.org/stable/24939139

Hong, K. C., Ling, T. C., Islam, S., Kolandaisamy, R. AL, Shibghatullah, A. S. Bin, Nidzam, N. N. B. S., Sarsam, S., & Zaman, H. B. (2025). Generative AI-Powered Synthetic Data for Enhancing Predictive Analytics in Blood Donation Supply Management: A Comparative Study of Machine Learning Models. *International Journal on Advanced Science, Engineering and Information Technology*, *15*(1), 9–19. https://doi.org/10.18517/IJASEIT.15.1.20498

IBM. (2025). *What Is Random Forest? | IBM*. IBM. https://www.ibm.com/think/topics/random-forest

interos.ai. (2025). *Supply Chain Resilience & Risk Management Software*. Interos.Ai. https://www.interos.ai/

Jackson, I., Ivanov, D., Dolgui, A., & Namdar, J. (2024). Generative artificial intelligence in supply chain and operations management: a capability-based framework for analysis and implementation. *International Journal of Production Research*, *62*(17), 6120–6145. https://doi.org/10.1080/00207543.2024.2309309

Jauhar, S. K., Harinath, S., Krishnaswamy, V., & Paul, S. K. (2024). Explainable artificial intelligence to improve the resilience of perishable product supply chains by leveraging customer characteristics. *Annals of Operations Research*, 1–40. https://doi.org/10.1007/S10479-024-06348-Z/TABLES/12

Jayasinghe, G. A. P. A. A., Prabodhika, A. P. K. J., & Wijayanayake, A. N. (2023). Risk Analysis on Green Supply Chain Management: A Systematic Literature Review. *ICARC 2023 - 3rd*

International Conference on Advanced Research in Computing: Digital Transformation for Sustainable Development, 280–285. https://doi.org/10.1109/ICARC57651.2023.10145646

Jia, F. (Jeff), Seuring, S., Chen, L., & Azadegan, A. (2024). Guest editorial: Supply chain transparency: opportunities, challenges and risks. *International Journal of Operations and Production Management*, *44*(9), 1525–1538. https://doi.org/10.1108/IJOPM-09-2024-992/FULL/PDF

Junejo, I., Sohu, J. M., Alwadi, B. M., Ejaz, F., Nasir, A., & Hossain, M. B. (2025). Green supply chain management and SMEs sustainable performance in developing country: role of green knowledge sharing, green innovation and big data-driven supply chain. *Discover Sustainability*, *6*(1), 1–19. https://doi.org/10.1007/S43621-025-01055-6/METRICS

Kang, P. S., & Bhawna, B. (2025). Enhancing supply chain resilience through supervised machine learning: supplier performance analysis and risk profiling for a multi-class classification problem. *Business Process Management Journal*, *ahead-of-print*(ahead-of-print). https://doi.org/10.1108/BPMJ-03-2024-0174/FULL/XML

Kassa, A., Kitaw, D., Stache, U., Beshah, B., & Degefu, G. (2023). Artificial intelligence techniques for enhancing supply chain resilience: A systematic literature review, holistic framework, and future research. *Computers & Industrial Engineering*, *186*, 109714. https://doi.org/10.1016/J.CIE.2023.109714

Kavlakoglu, E., & Russi, E. (2025). *What is XGBoost? | IBM*. IBM. https://www.ibm.com/think/topics/xgboost

Khan, M., Ajmal, M. M., Jabeen, F., Talwar, S., & Dhir, A. (2023). Green supply chain management in manufacturing firms: A resource-based viewpoint. *Business Strategy and the Environment*, *32*(4), 1603–1618. https://doi.org/10.1002/BSE.3207;CTYPE:STRING:JOURNAL

Kholaif, M. M. N. H. K., Gu, W., Tang, X., & Salama, W. H. (2024). Post-emergencies opportunities for green supply chain management and urban supply chains' robustness; the moderating influence of blockchains as a sustainable urban resources management technique. *Journal of Cleaner Production*, *475*, 143646. https://doi.org/10.1016/J.JCLEPRO.2024.143646

Khulud, K., Masudin, I., Zulfikarijah, F., Restuputri, D. P., & Haris, A. (2023). Sustainable Supplier Selection through Multi-Criteria Decision Making (MCDM) Approach: A Bibliometric Analysis. *Logistics 2023, Vol. 7, Page 96*, *7*(4), 96. https://doi.org/10.3390/LOGISTICS7040096

Kingma, D. P., & Welling, M. (2013). Auto-Encoding Variational Bayes. *2nd International Conference on Learning Representations, ICLR 2014 - Conference Track Proceedings*, *1*. https://doi.org/10.61603/ceas.v2i1.33

Lin, C.-Y., Alam, S. S., Ho, Y.-H., Al-Shaikh, M. E., & Sultan, P. (2020). Adoption of Green Supply Chain Management among SMEs in Malaysia. *Sustainability*, *12*(16), 6454. https://doi.org/10.3390/su12166454

Liu, Y., He, B., Hildebrandt, M., Buchner, M., Inzko, D., Wernert, R., Weigel, E., Beyer, D., Berbalk, M., & Tresp, V. (2023). A Knowledge Graph Perspective on Supply Chain Resilience. *Machine Learning*. http://ceur-ws.org

Long, Y., Kroeger, S., Zaeh, M. F., & Brintrup, A. (2025). Leveraging synthetic data to tackle machine learning challenges in supply chains: challenges, methods, applications, and research opportunities. *International Journal of Production Research*. https://doi.org/10.1080/00207543.2024.2447927;PAGE:STRING:ARTICLE/CHAPTER

Lundberg, S. M., & Lee, S. I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems*, *2017-December*, 4766–4775. https://arxiv.org/pdf/1705.07874

MacQueen, J. B. (1967). Some methods for classification and analysis of multivariate observations. In L. M. Cam & J. Neyman (Eds.), *Proceedings of the fifth Berkeley symposium on mathematical statistics and probability (Vol. 1, pp. 281–297)*. University of California Press.

Majumdar, A., S, J., Kaliyan, M., & Agrawal, R. (2023). Selection of resilient suppliers in manufacturing industries post-COVID-19: implications for economic and social sustainability in emerging economies. *International Journal of Emerging Markets*, *18*(10), 3657–3675. https://doi.org/10.1108/IJOEM-09-2021-1393

McCulloch, W. S., & Pitts, W. (1943). A logical calculus of the ideas immanent in nervous activity. *The Bulletin of Mathematical Biophysics*, *5*(4), 115–133. https://doi.org/10.1007/BF02478259

Miletic, M., & Sariyar, M. (2024). Challenges of Using Synthetic Data Generation Methods for Tabular Microdata. *Applied Sciences 2024, Vol. 14, Page 5975*, *14*(14), 5975. https://doi.org/10.3390/APP14145975

Nădăban, S., Dzitac, S., & Dzitac, I. (2016). Fuzzy TOPSIS: A General View. *Procedia Computer Science*, *91*, 823–831. https://doi.org/https://doi.org/10.1016/j.procs.2016.07.088

Nezianya, M. C., Adebayo, A. O., & Ezeliora, P. (2024). A critical review of machine learning applications in supply chain risk management. *World Journal of Advanced Research and Reviews*, *23*(3), 1554–1567. https://doi.org/10.30574/WJARR.2024.23.3.2760

Pecan AI. (2025). *Predictive Analytics Software For Humans*. Pecan AI. https://www.pecan.ai/

Placek, M. (2024). *Adoption of cutting-edge technologies by supply chain companies in 2021*. Statista.

Probst, P., Wright, M. N., & Boulesteix, A. L. (2019). Hyperparameters and tuning strategies for random forest. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, *9*(3), e1301. https://doi.org/10.1002/WIDM.1301

QIMA. (2025). *QIMA Signals: AI Tool for Supply Chain Risk Management* . QIMA. https://www.qima.com/consumer-products/esg-solutions/risk-mitigation-solutions/qima-signals

Rahiminia, M., Razmi, J., Shahrabi Farahani, S., & Sabbaghnia, A. (2023). Cluster-based supplier segmentation: a sustainable data-driven approach. *Modern Supply Chain Research and Applications*, *5*(3), 209–228. https://doi.org/10.1108/MSCRA-05-2023-0017

Rosenblatt, F. (1958). The perceptron: A probabilistic model for information storage and organization in the brain. *Psychological Review*, *65*(6), 386–408. https://doi.org/10.1037/h0042519

Saaty, R. W. (1987). The analytic hierarchy process—what it is and how it is used. *Mathematical Modelling*, *9*(3–5), 161–176. https://doi.org/10.1016/0270-0255(87)90473-8

Saaty, T. L. (2008). Decision making with the analytic hierarchy process. *Int. J. Services Sciences*, *1*(1), 83–98. https://www.rafikulislam.com/uploads/resourses/197245512559a37aadea6d.pdf

Sander, J., Ester, M., Kriegel, H.-P., & Xu, X. (1998). Density-Based Clustering in Spatial Databases: The Algorithm GDBSCAN and Its Applications. *Data Mining and Knowledge Discovery*, *2*(2), 169–194. https://doi.org/10.1023/A:1009745219419

SAP. (2025). *Supply Chain Management (SCM) Software Solutions*. SAP. https://www.sap.com/sea/products/scm.html

Shapley, L. S. (1953). A Value for n-Person Games. In *Contributions to the Theory of Games (AM-28), Volume II* (pp. 307–318). Princeton University Press. https://doi.org/10.1515/9781400881970-018

Shawon, R. E. R., Hasan, M. R., Rahman, M. A., Ghandri, M., Lamari, I. A., Kawsar, M., & Akter, R. (2025). Designing and Deploying AI Models for Sustainable Logistics Optimization: A Case Study on Eco-Efficient Supply Chains in the USA. *Journal of Ecohumanism*, *4*(2), 2143–2166. https://doi.org/10.62754/joe.v4i2.6610

Singh, K., Chaudhuri, R., & Chatterjee, S. (2025). Assessing the impact of digital transformation on green supply chain for achieving carbon neutrality and accelerating circular economy initiatives. *Computers & Industrial Engineering*, *201*, 110943. https://doi.org/10.1016/J.CIE.2025.110943

S&P Global. (2025). *Corporate Responsibility | S&P Global*. S&P Global. https://www.spglobal.com/en/who-we-are/corporate-responsibility

Stanton, J. M. (2001). Galton, Pearson, and the Peas: A Brief History of Linear Regression for Statistics Instructors. *Journal of Statistics Education*, *9*(3). https://doi.org/10.1080/10691898.2001.11910537

Statista Research Department. (2024). *Sustainability reporting rates of firms worldwide from 2011 to 2024, by region*. Statista.

Sujay Vailshery, L. (2024, November 26). *IT sustainability drivers 2023| Statista*. Statista. https://www-statista-com.tarc.idm.oclc.org/statistics/1385351/it-sustainability-drivers/

Sutrisno, A., & Kumar, V. (2023). Supply chain sustainability risk assessment model using integration of the preference selection index (PSI) and the Shannon entropy. *International Journal of Quality and Reliability Management*, *40*(3), 674–708. https://doi.org/10.1108/IJQRM-06-2021-0191/FULL/PDF

Talekar, B. (2020). A Detailed Review on Decision Tree and Random Forest. *Bioscience Biotechnology Research Communications*, *13*(14), 245–248. https://doi.org/10.21786/bbrc/13.14/57

Thormundsson, B. (2023). *Artificial intelligence (AI) adoption rate in supply chain and manufacturing businesses worldwide in 2022 and 2025*. Statista.

Tian, G., Lu, W., Zhang, X., Zhan, M., Dulebenets, M. A., Aleksandrov, A., Fathollahi-Fard, A. M., & Ivanov, M. (2023). A survey of multi-criteria decision-making techniques for green logistics and low-carbon transportation systems. *Environmental Science and Pollution Research 2023 30:20*, *30*(20), 57279–57301. https://doi.org/10.1007/S11356-023-26577-2

Tronnebati, I., Jawab, F., Frichi, Y., & Arif, J. (2024). Green Supplier Selection Using Fuzzy AHP, Fuzzy TOPSIS, and Fuzzy WASPAS: A Case Study of the Moroccan Automotive Industry. *Sustainability 2024, Vol. 16, Page 4580*, *16*(11), 4580. https://doi.org/10.3390/SU16114580

United States Environmental Protection Agency. (2025, June 27). *Random Forest Modeling | US EPA*. United States Environmental Protection Agency. https://www.epa.gov/streamflow-duration-assessment/random-forest-modeling

Varchandi, S., Memari, A., & Jokar, M. R. A. (2024). An integrated best–worst method and fuzzy TOPSIS for resilient-sustainable supplier selection. *Decision Analytics Journal*, *11*, 100488. https://doi.org/10.1016/j.dajour.2024.100488

Werbos, P. J. (1990). Backpropagation Through Time: What It Does and How to Do It. *Proceedings of the IEEE*, *78*(10), 1550–1560. https://doi.org/10.1109/5.58337

Whieldon, E., Ballerini, N., Costello, L., Ngo, C. H., MacFarland, M., & Staeheli, M. (2024). *Assessing the missing links in sustainable supply chain management*. S&P Global.

Wu, T., & Zuo, M. (2023). Green supply chain transformation and emission reduction based on machine learning. *Science Progress*, *106*(1), 00368504231165679. https://doi.org/10.1177/00368504231165679

Xu, L., Skoularidou, M., Cuesta-Infante, A., & Veeramachaneni, K. (2019). Modeling tabular data using conditional GAN. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems*. Curran Associates Inc.

Yang, M., Lim, M. K., Qu, Y., Ni, D., & Xiao, Z. (2023). Supply chain risk management with machine learning technology: A literature review and future research directions. *Computers & Industrial Engineering*, *175*, 108859. https://doi.org/10.1016/J.CIE.2022.108859

Yang, X., Su, M., Zhang, T., & Jia, F. (2024). The integration of sustainability into the service supply chain: towards a research agenda. *International Journal of Logistics Research and Applications*. https://doi.org/10.1080/13675567.2024.2391978

Zejjari, I., & Benhayoun, I. (2024). The use of artificial intelligence to advance sustainable supply chain: retrospective and future avenues explored through bibliometric analysis. *Discover Sustainability*, *5*(1), 1–14. https://doi.org/10.1007/S43621-024-00364-6/FIGURES/5

Zhao, C., Zhang, M., Bai, J., Wu, J., & Chang, I. S. (2025). A review of the application of machine learning in carbon emission assessment studies: prediction optimization and driving factor selection. *Science of The Total Environment*, *987*, 179678. https://doi.org/10.1016/J.SCITOTENV.2025.179678

Zulqarnain, R. M., Dai, H.-L., Ma, W.-X., Siddique, I., Askar, S., & Naveed, H. (2024). Supplier selection in green supply chain management using correlation-based TOPSIS in a q-rung orthopair fuzzy soft environment. *Heliyon*, *10*(11), e32145. https://doi.org/10.1016/j.heliyon.2024.e32145

Zyoud, S. H., & Fuchs-Hanusch, D. (2017). A bibliometric-based survey on AHP and TOPSIS techniques. *Expert Systems with Applications*, *78*, 158–181. https://doi.org/10.1016/J.ESWA.2017.02.016